BONES OF THE UPPER LIMB

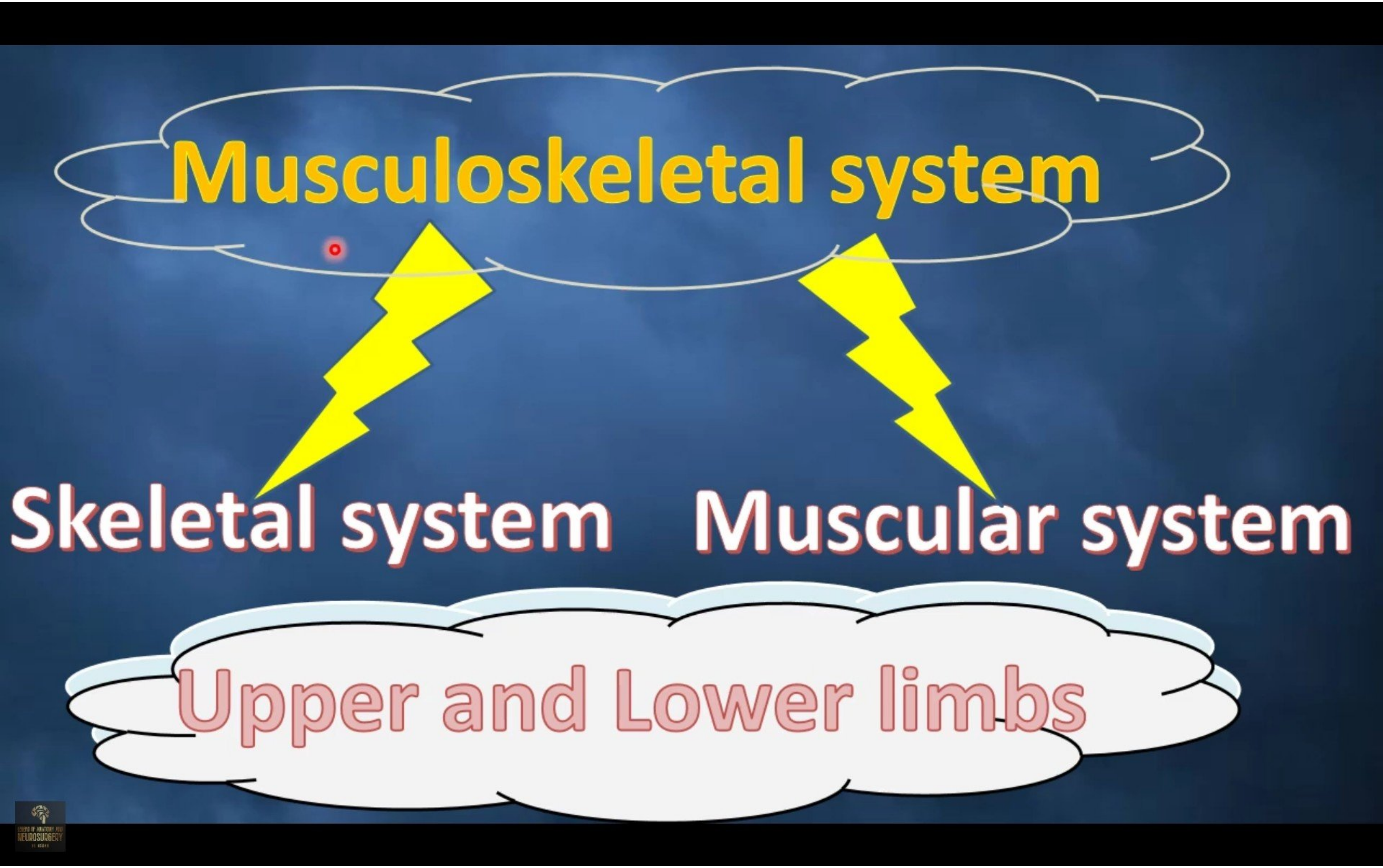
Department of Anatomy

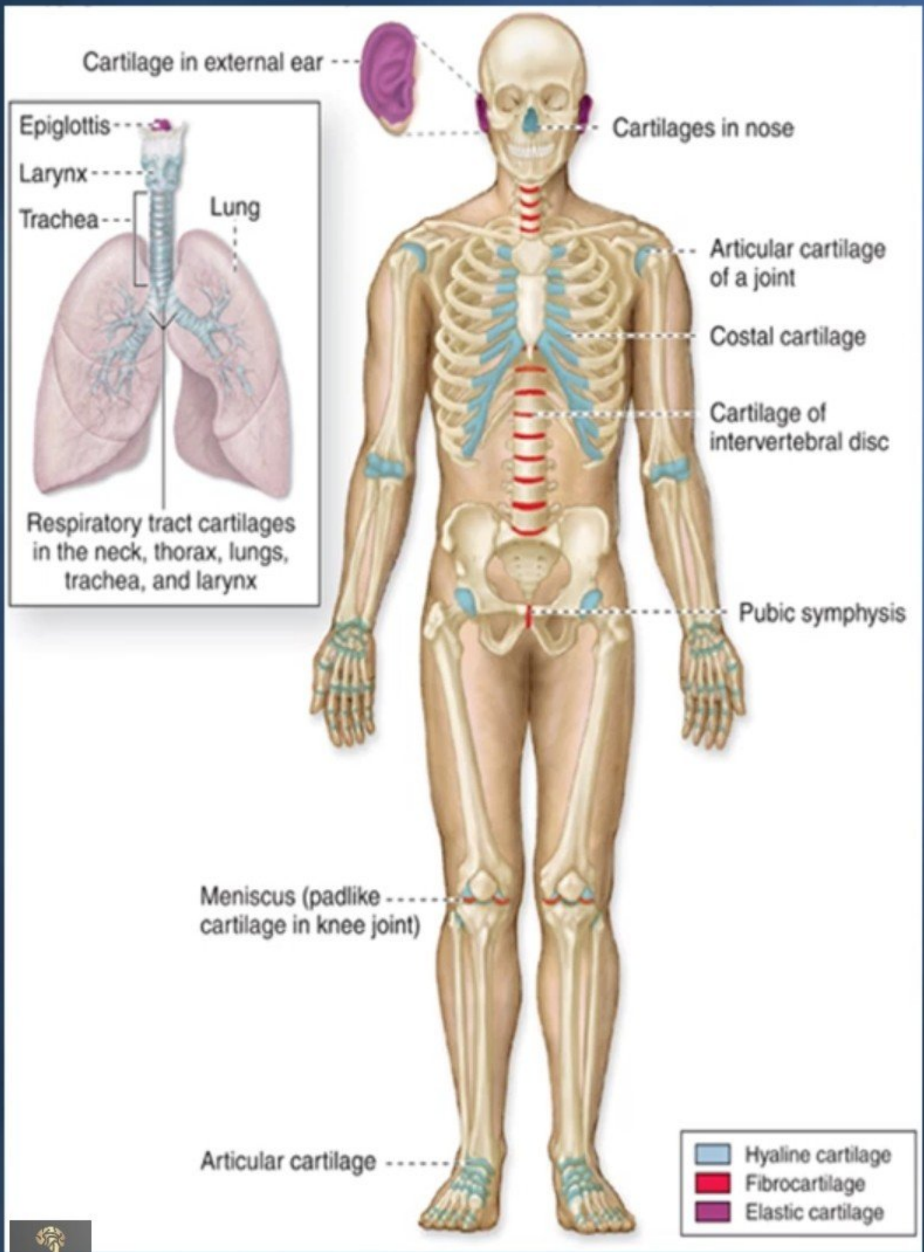
Musculoskeletal System

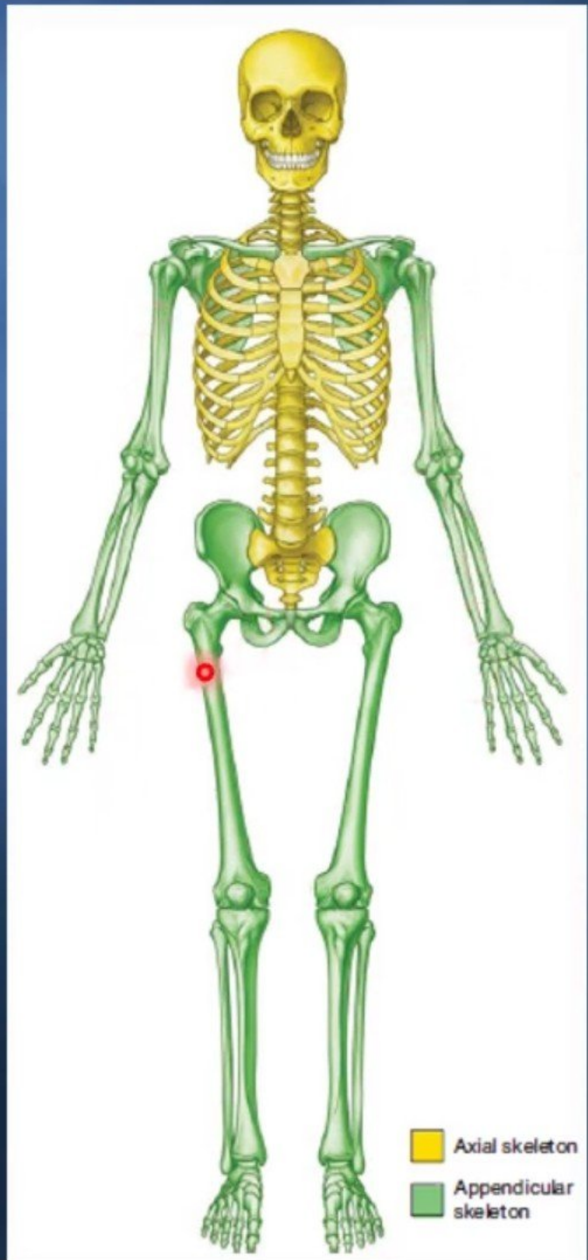
The Upper Limb (1)

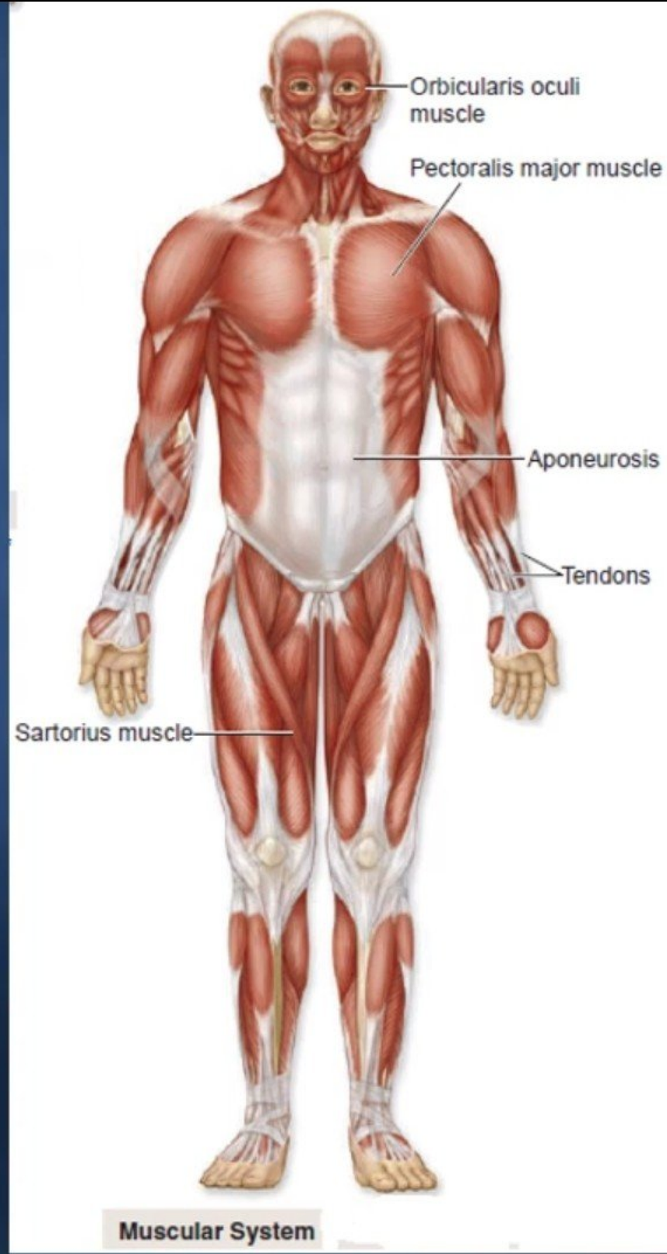
First Part

Dr. Osman Ahmed Osman


Musculoskeletal system
Skeletal system
Muscular system
Upper and Lower limbs


Cartilage in external ear
Epiglottis
Larynx
Trachea
Lung
Respiratory tract cartilages in the neck, thorax, lungs, trachea, and larynx
Cartilages in nose
Articular cartilage of a joint
Costal cartilage
Cartilage of intervertebral disc
Pubic symphysis
Meniscus (padlike cartilage in knee joint)
Articular cartilage
Hyaline cartilage
Fibrocartilage
Elastic cartilage


Axial skeleton
Appendicular skeleton


Orbicularis oculi muscle
Pectoralis major muscle
Aponeurosis
Tendons
Sartorius muscle
Muscular System

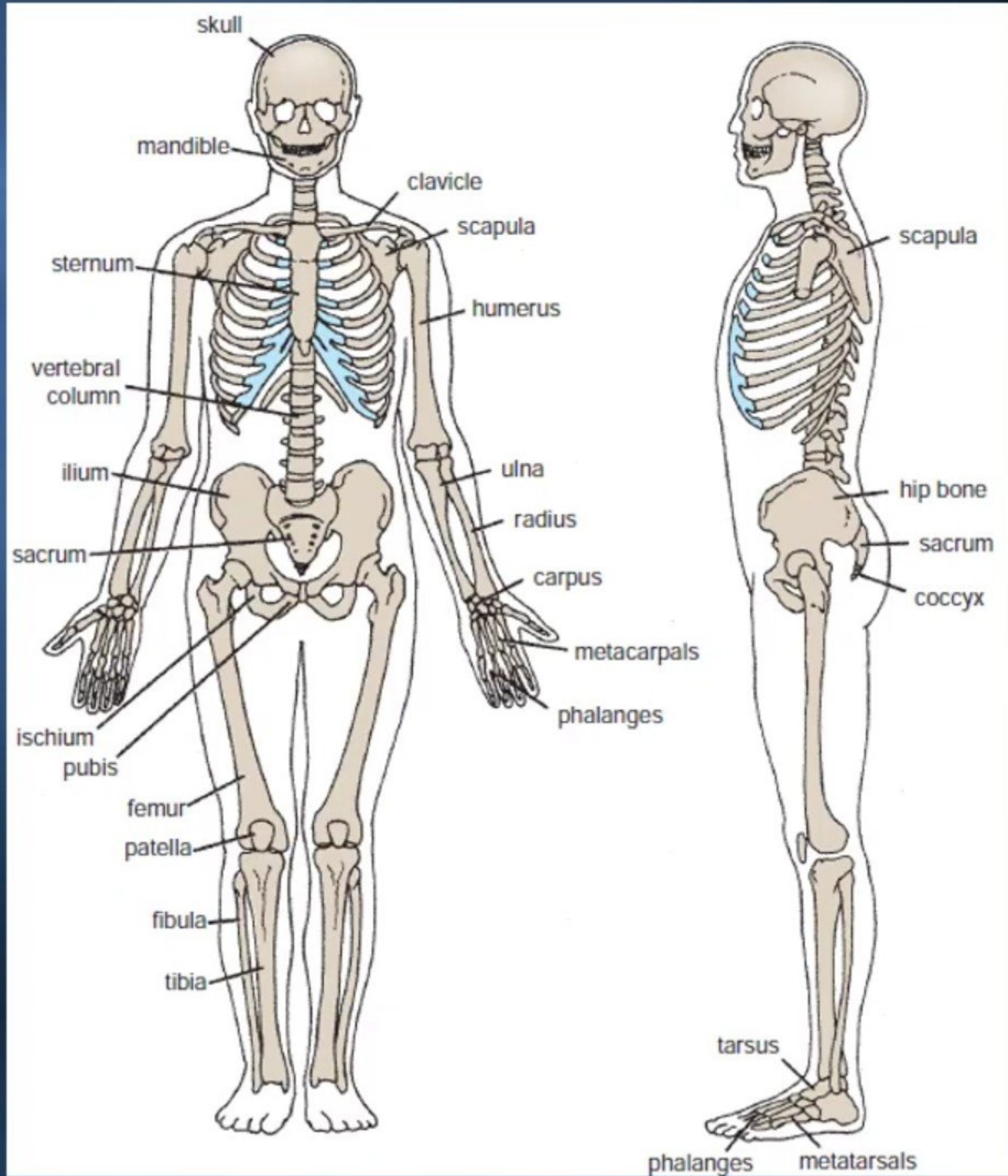
Lecturer Points

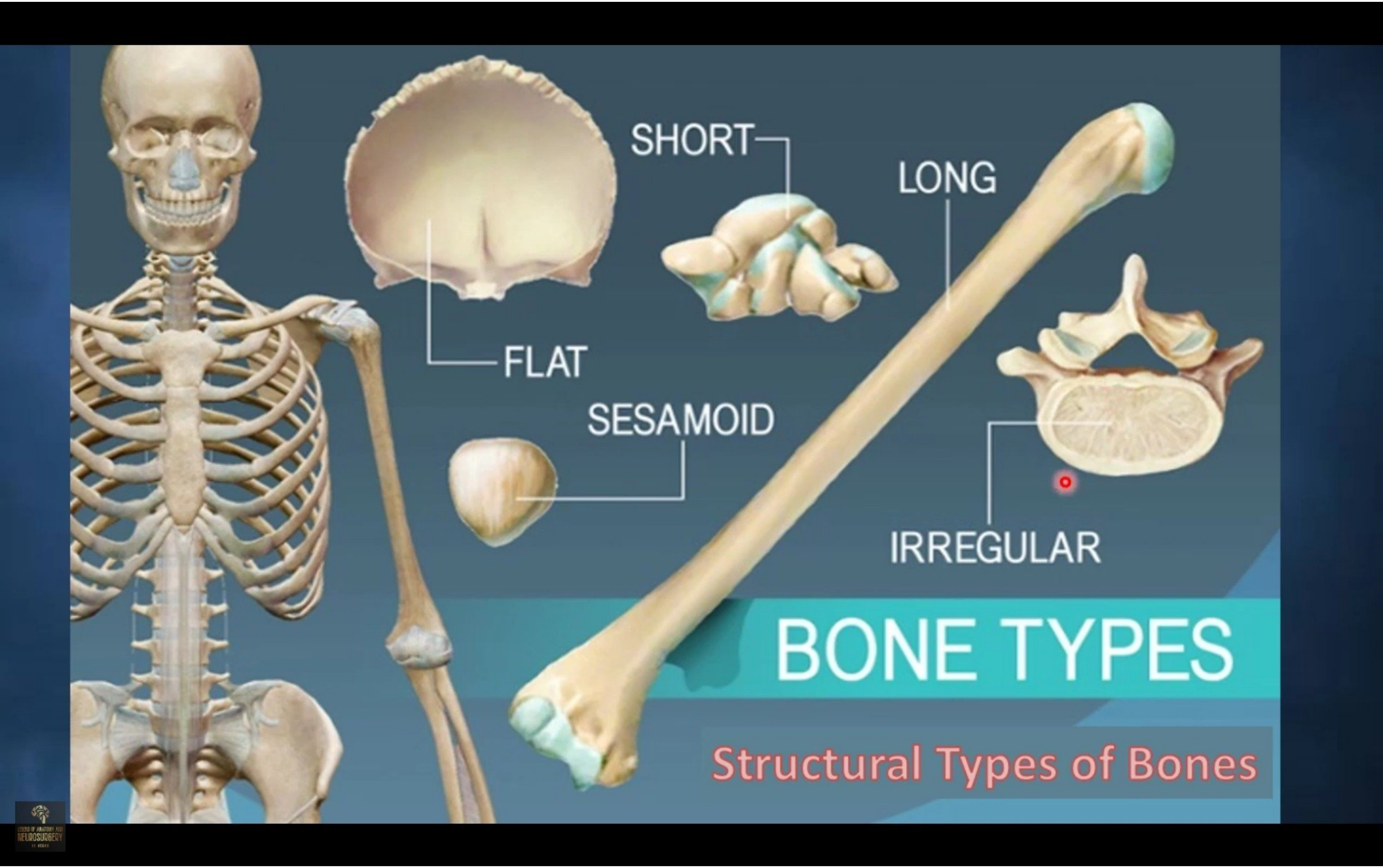
- **Skeletal System Parts.**
- **Structural Types of Bones**
- JOINTS
  - Definition.
  - Type of joints.
  - Typical synovial joint .
  - The accessory structures associated with synovial joint and major features.
  - Types of synovial joint according the shape.
  - Type of Cartilaginous joints:
- Skelton of Musculoskeletal system.
- The Upper Limbs
- *The Bones of Upper Limb are:*
- *Pectoral or Shoulder Girdle*
- Clavicle
  - Articulation of the clavicle
  - Muscle Origins and Insertions and Ligament attachments
  - Normal X-ray
  - Fractures of the Clavicle
- Scapula.
  - Fracture of the Scapula
  - Articulation of head of humorous with scapula
  - Humerus
    - Fractures of Surgical Neck and Mid shaft Humerus
    - Supracondylar Fractures.
    - Nerves affected in Fractures of Humerus
- Bones of The Forearm
  - Ulna
  - Radius
  - Normal X-ray Radiographic Appearances
  - Fractures of radius & ulna
  - fractured elbow
- WRI ST (CARPUS)
- Carpal Bones
- Metacarpus
- Digits (Phalanges).

Skeletal System

❖ The human skeleton is divided into:

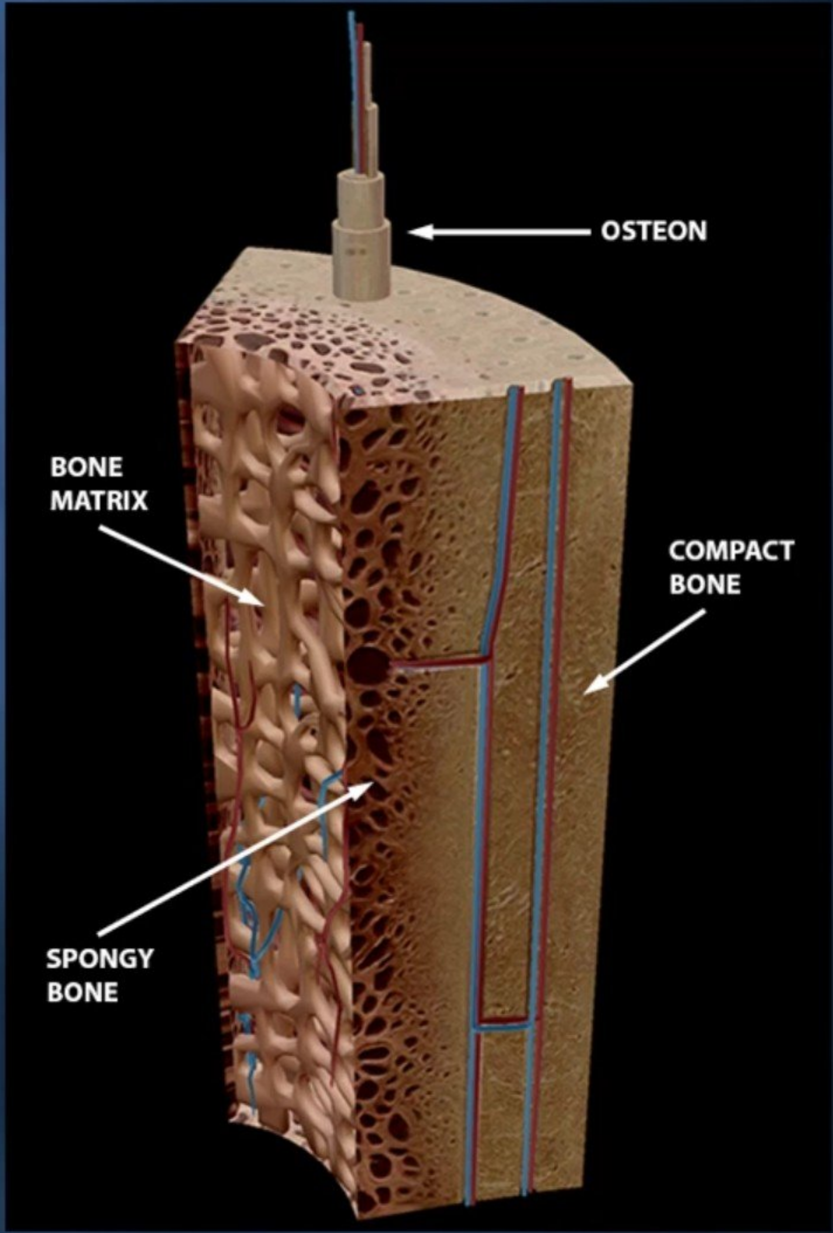
1. The **axial skeleton** is composed of the bones along the central axis of the body,
   A. The skull.
   B. The vertebral column.
   C. The thoracic cage.
2. The **appendicular skeleton** consists of the bones of the appendages .
   - Upper limbs.
   - lower limbs.

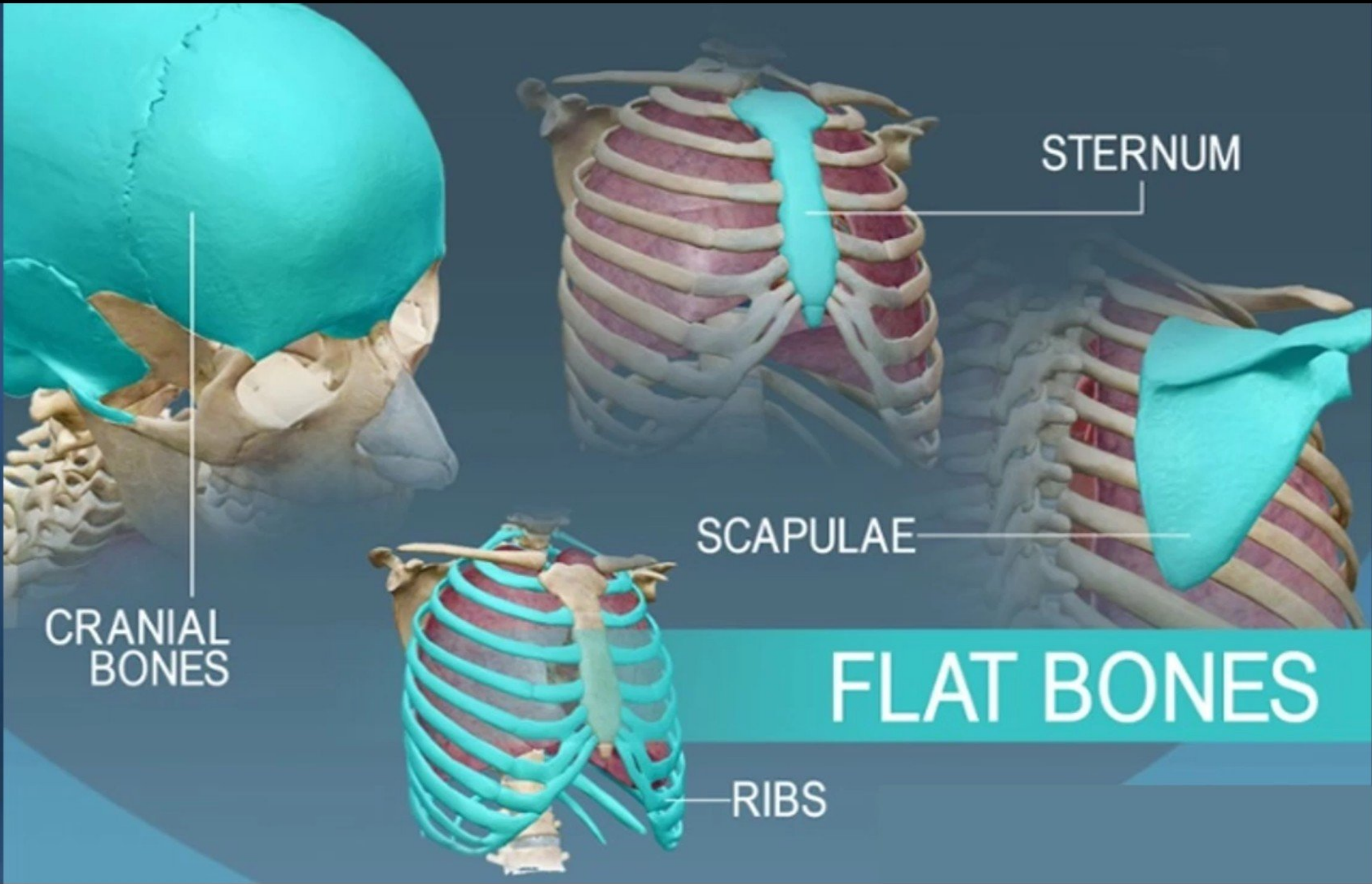


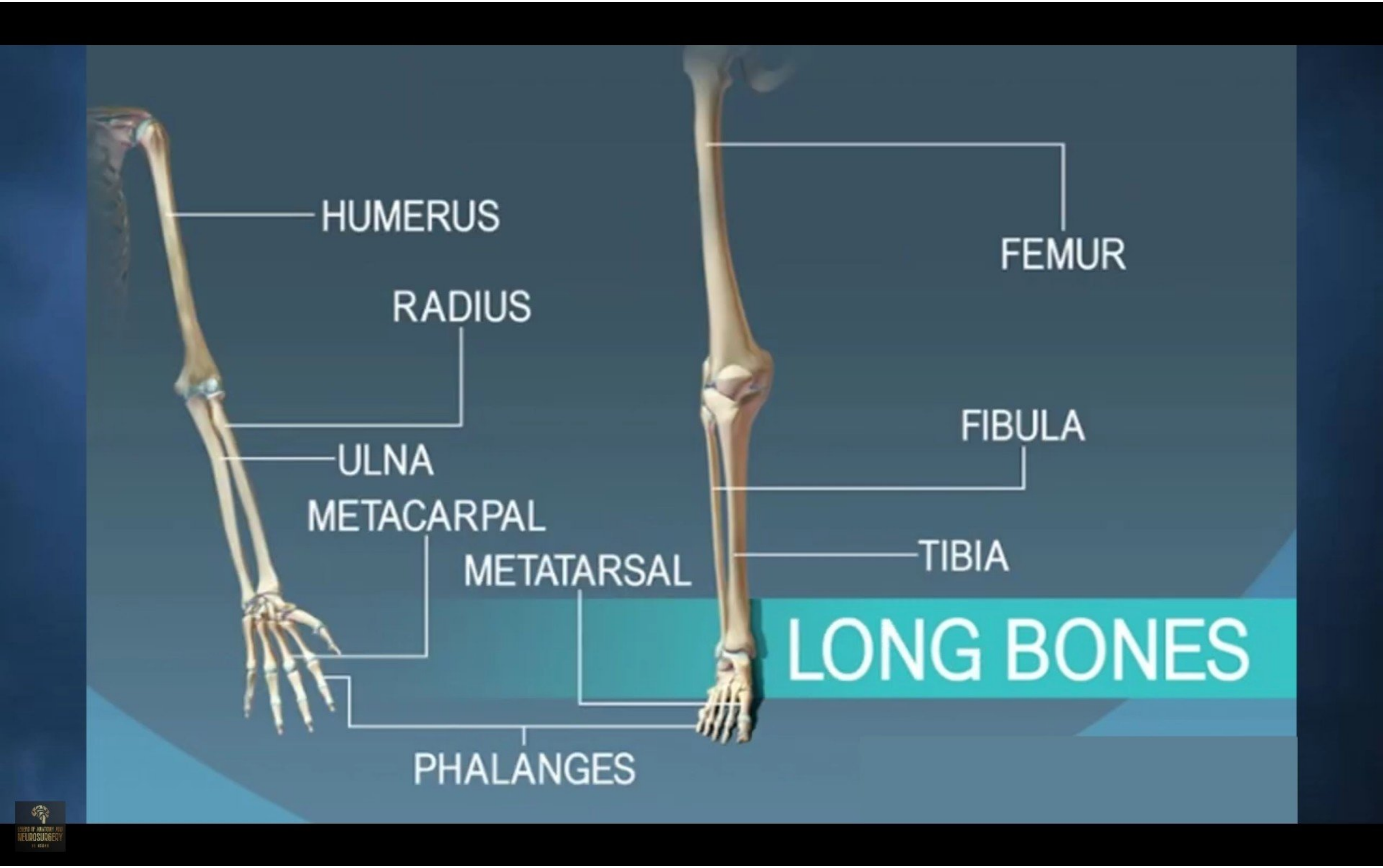

SHORT
LONG
FLAT
SESAMOID
IRREGULAR
BONE TYPES
Structural Types of Bones

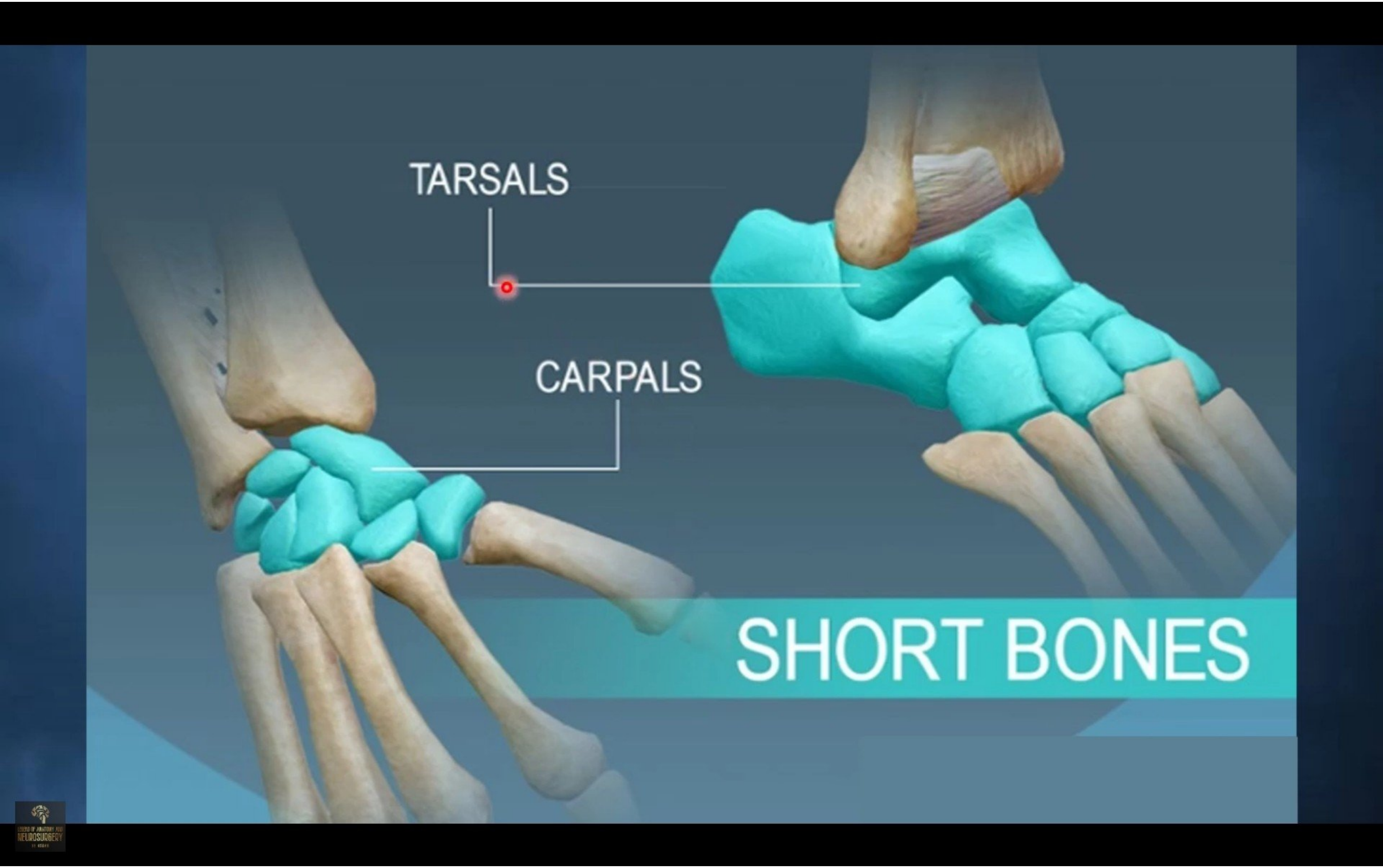
- The human skeleton has a number of functions, such as **protection and supporting weight**.
- Different types of bones have differing shapes related to their particular function.
- Types of bone on the basis of shape:

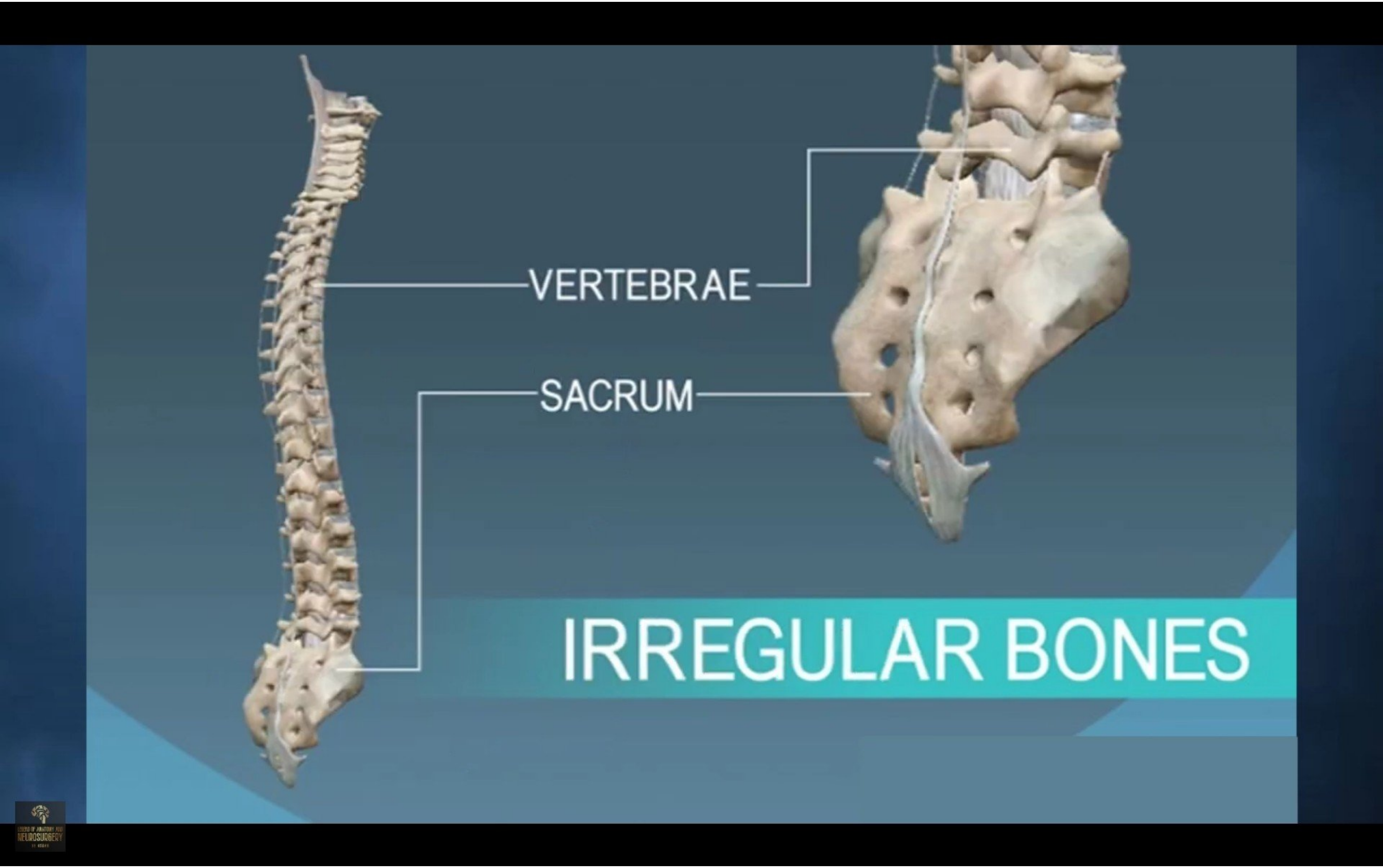
1. **Flat.**
2. **Long.**
3. **Short.**
4. **Irregular.**
5. **Sesamoid.**

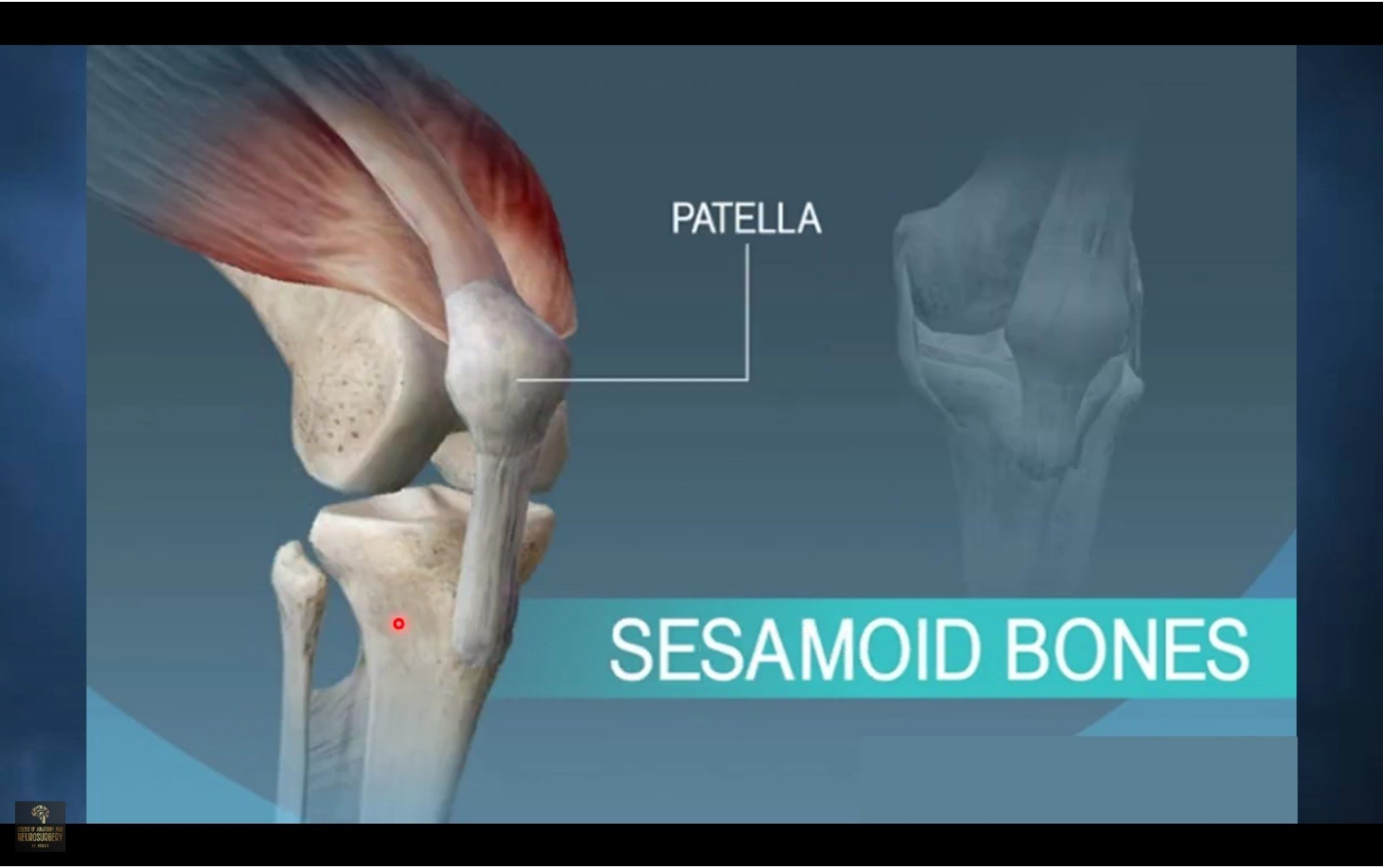



STERNUM
SCAPULAE
CRANIAL BONES
FLAT BONES
RIBS


HUMERUS
RADIUS
ULNA
METACARPAL
METATARSAL
PHALANGES
FEMUR
FIBULA
TIBIA
LONG BONES

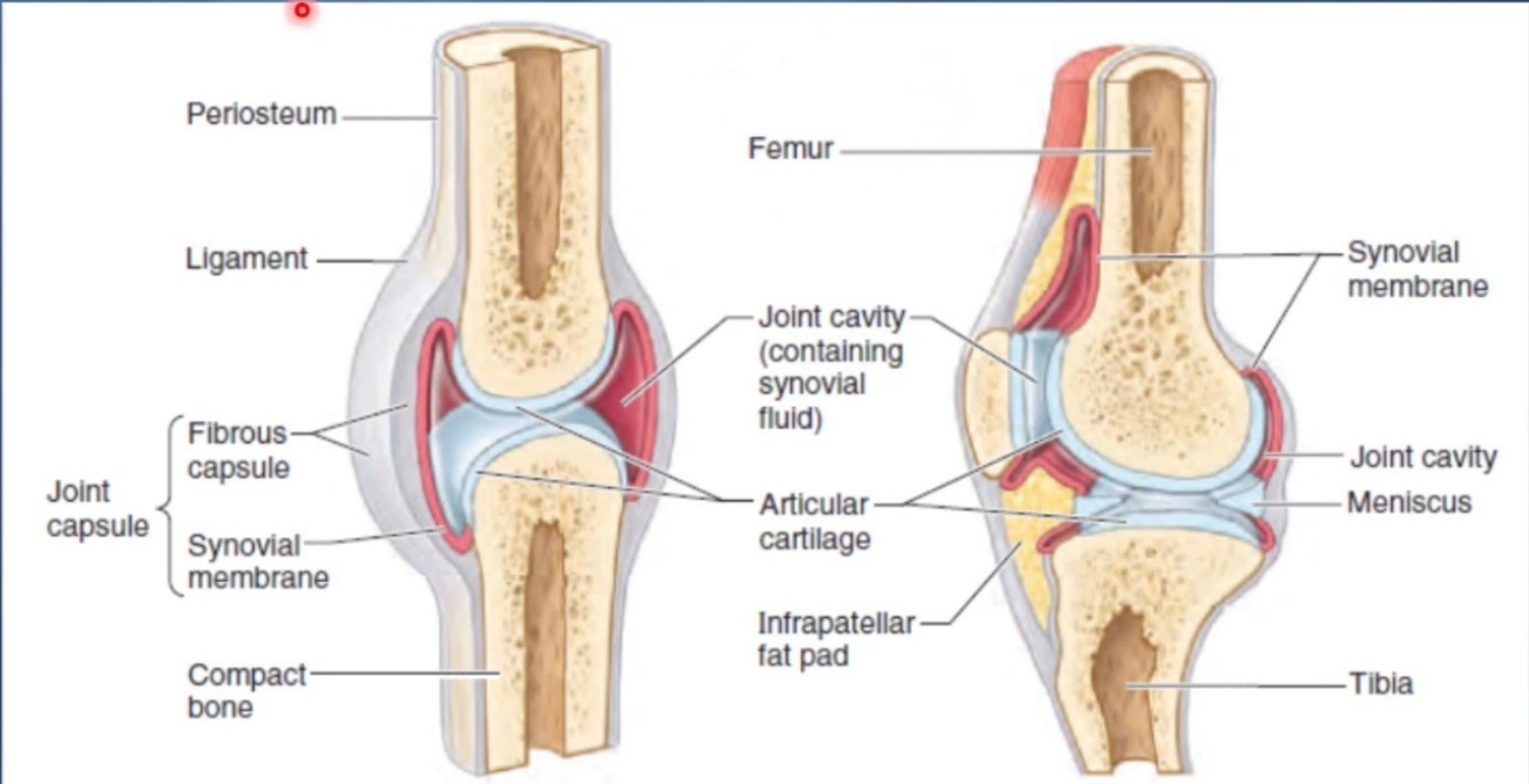

TARSALS
CARPALS
SHORT BONES


VERTEBRAE
SACRUM
IRREGULAR BONES


PATELLA
SESAMOID BONES

JOINTS

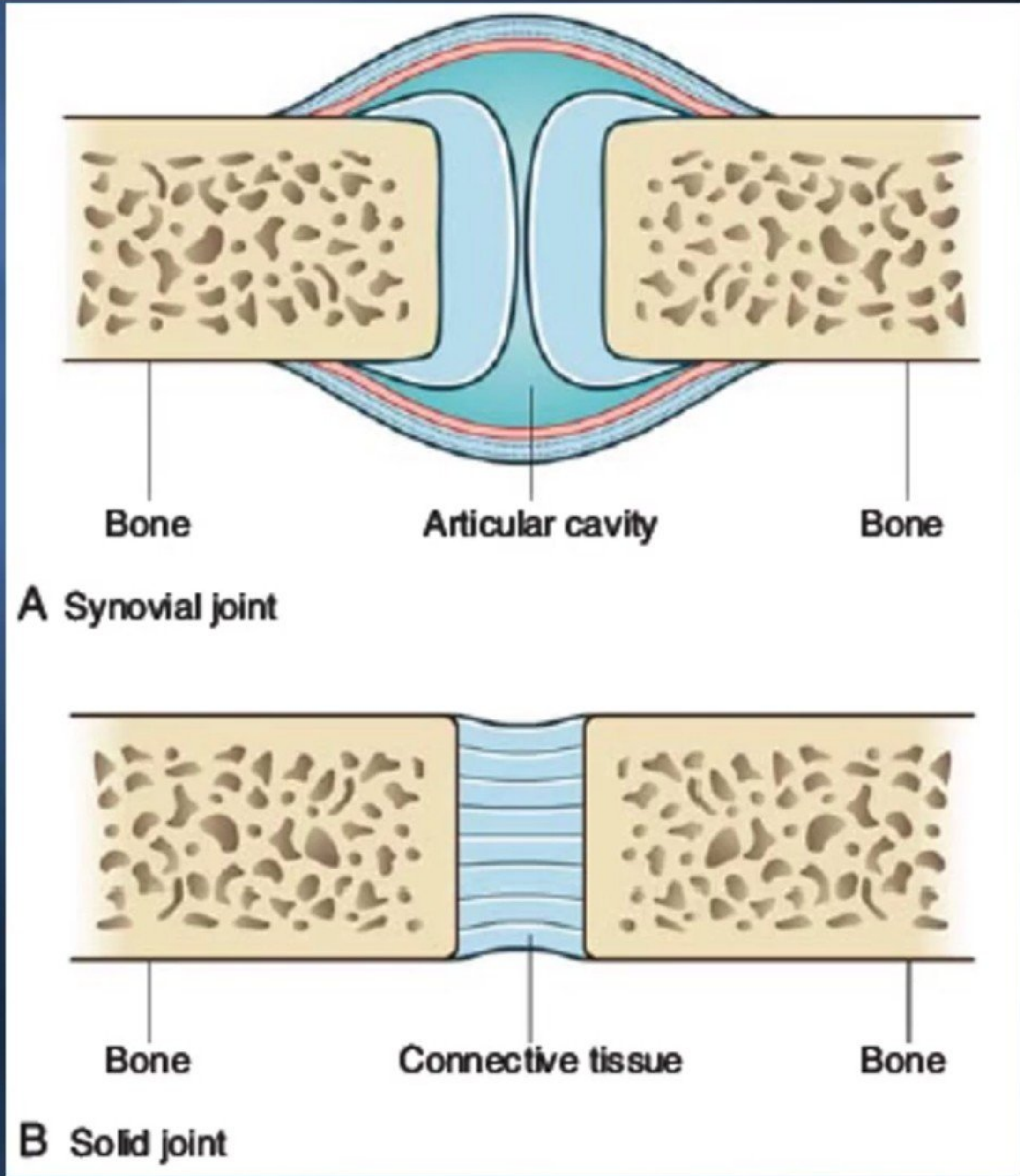
Joints (articulations) are sites where two or more bones meet.



(A) Synovial joint Schematic model Knee joint

Type of joints

- **Synovial joints:**
  - ✓ Are connection between skeletal component where the elements involved are seperted by narrow **articular cavity.**
- **Solid joints :**
  - ✓ Are connections between skeletal elements where the adjacent surfaces are liked together by **Fibrous connective tissue** or by **cartilage**, (usually **fibrocartilage**).
  - ✓ **No Cavity**.
  - ✓ **Movements** at this joint are **more restricted** than at synovial joints.

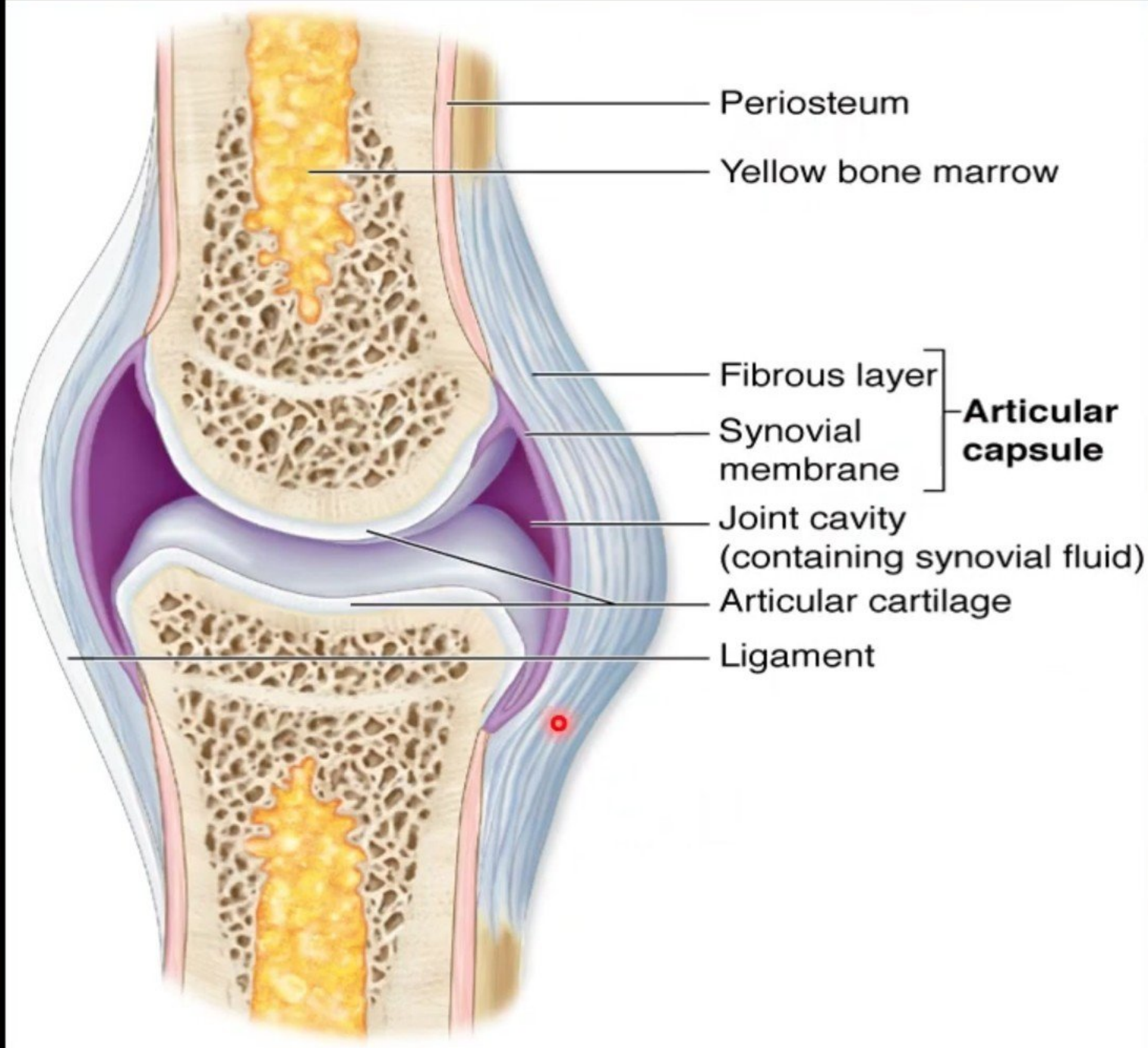


A Synovial joint

B Solid joint

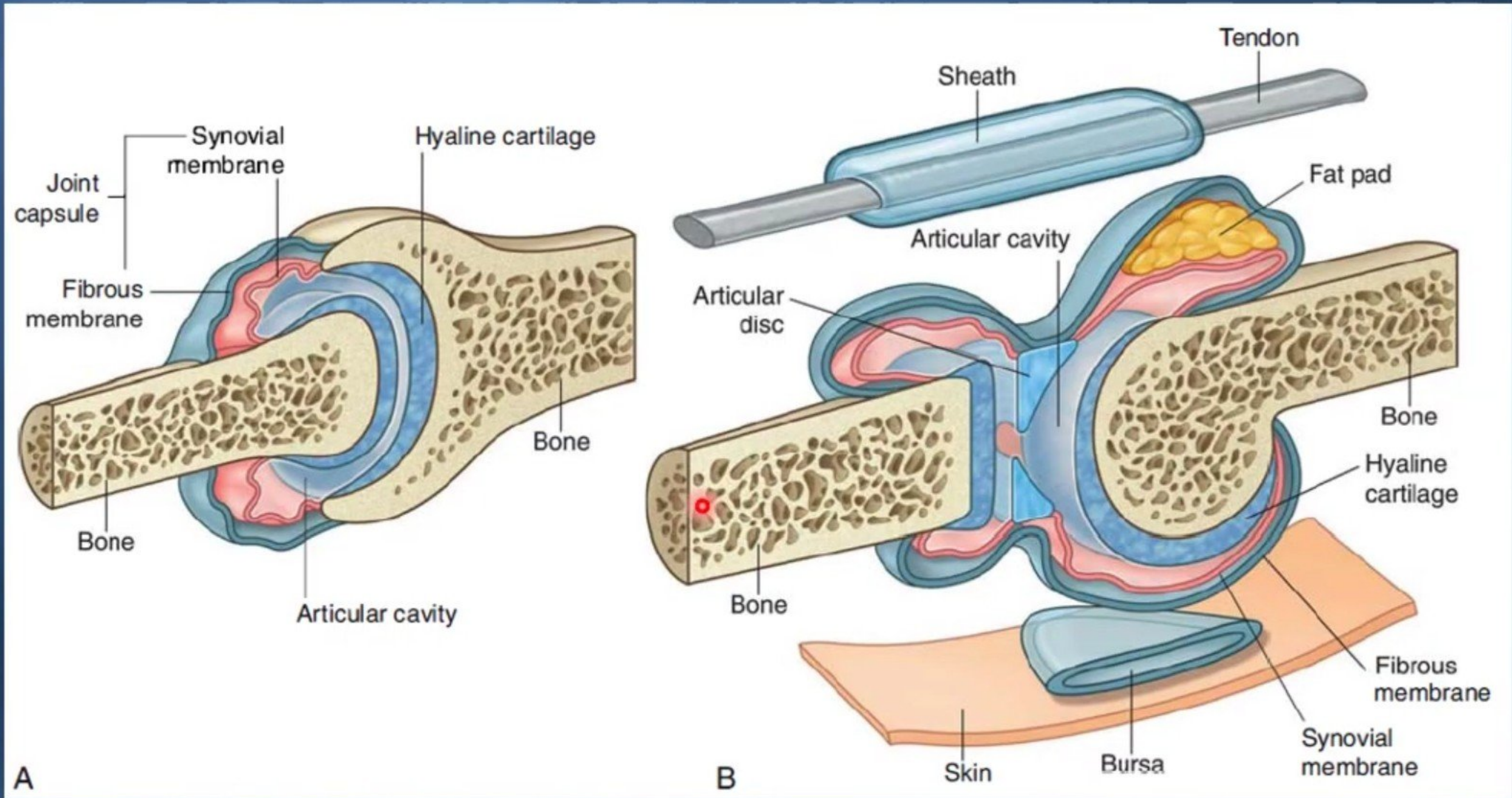
Typical synovial joint

- **Articular Capsule:**
  - **Fibrous layer.**
  - **Synovial membrane.**
- **Joint Cavity** (containing of Synovial fluid).
- Articular cartilage.
- Ligament.

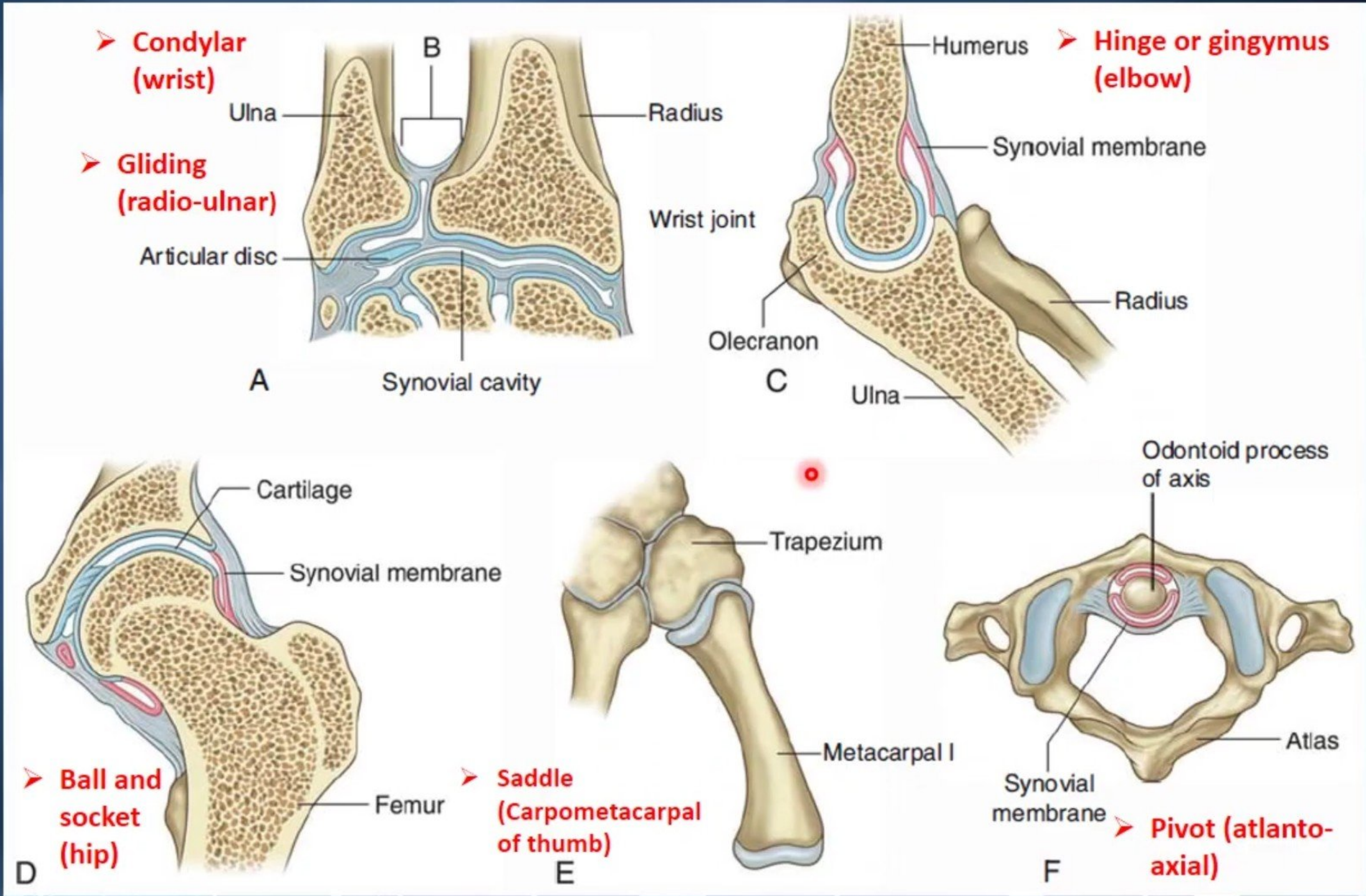


a Typical synovial joint

The accessory structures associated with synovial joint and major features.



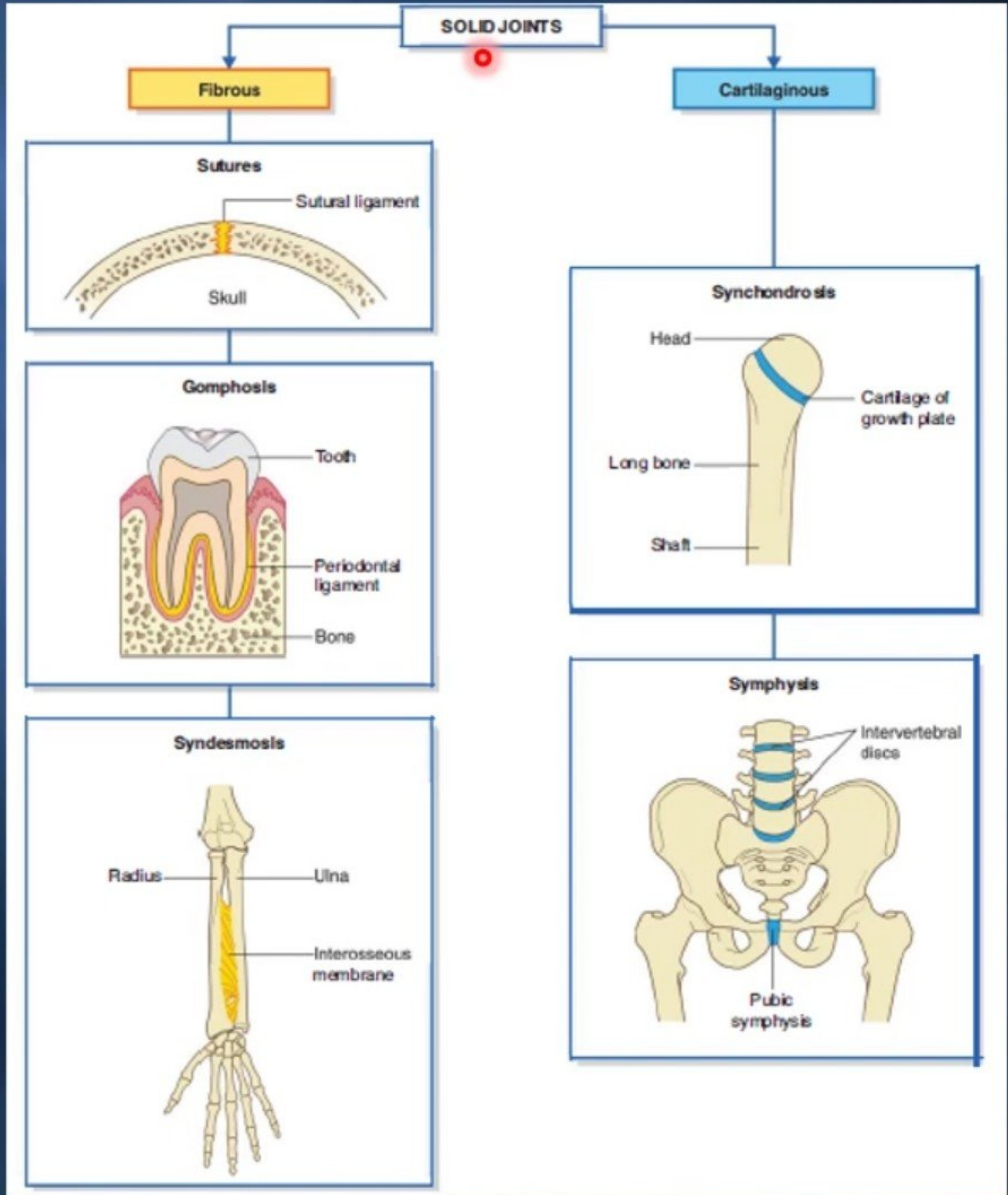
Types of synovial joint according the shape



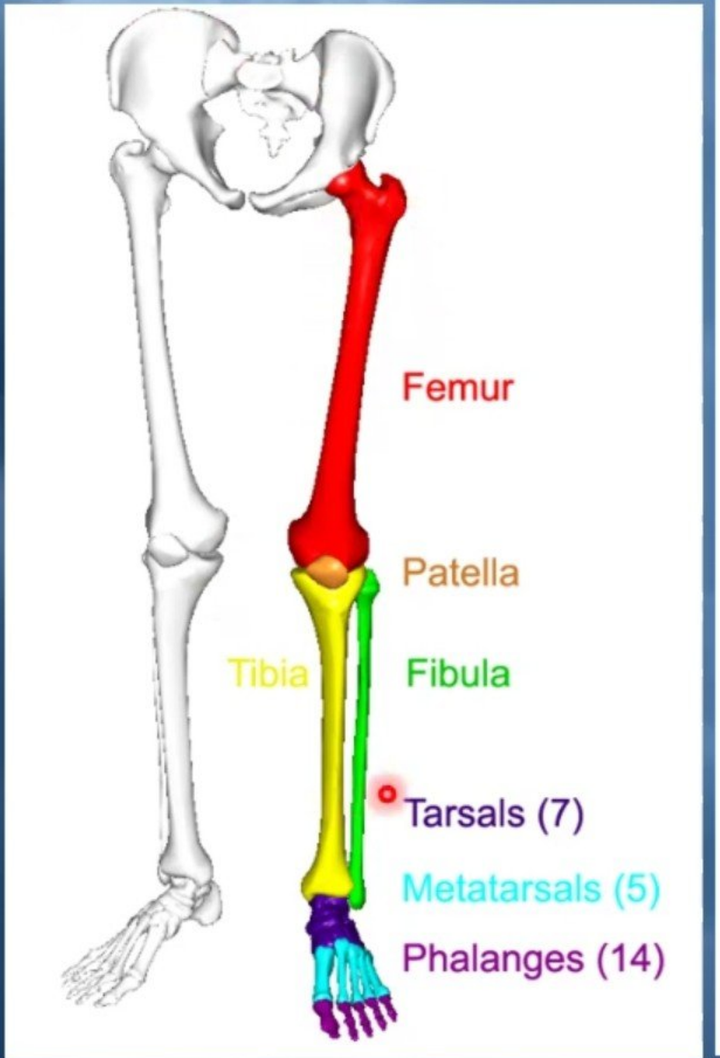
Type of Cartilaginous joints:

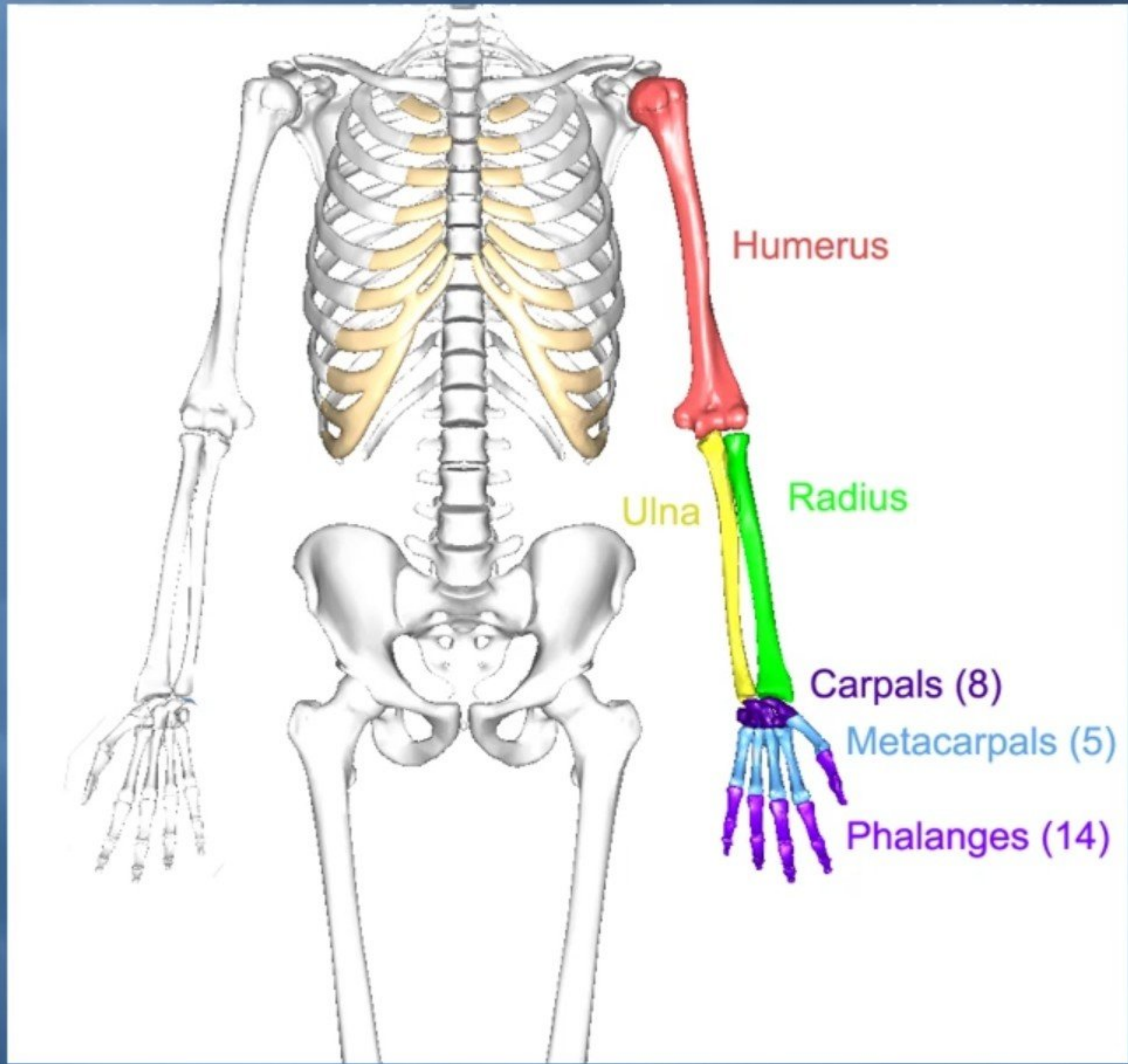
Synchondrosis occur where **two ossification centers** in a developing bone remain **separated by layer of cartilage**, for example the **growth palate** that occurs between **the head and shaft** of developing **long bones**.

Symphyses occur where two **separate bones** are **interconnected by cartilage**. Most of these types of joints occur **in midline** and include the **pubic symphysis** between the two pelvic bones and **intervertebral discs**.



Skelton of Musculoskeletal system

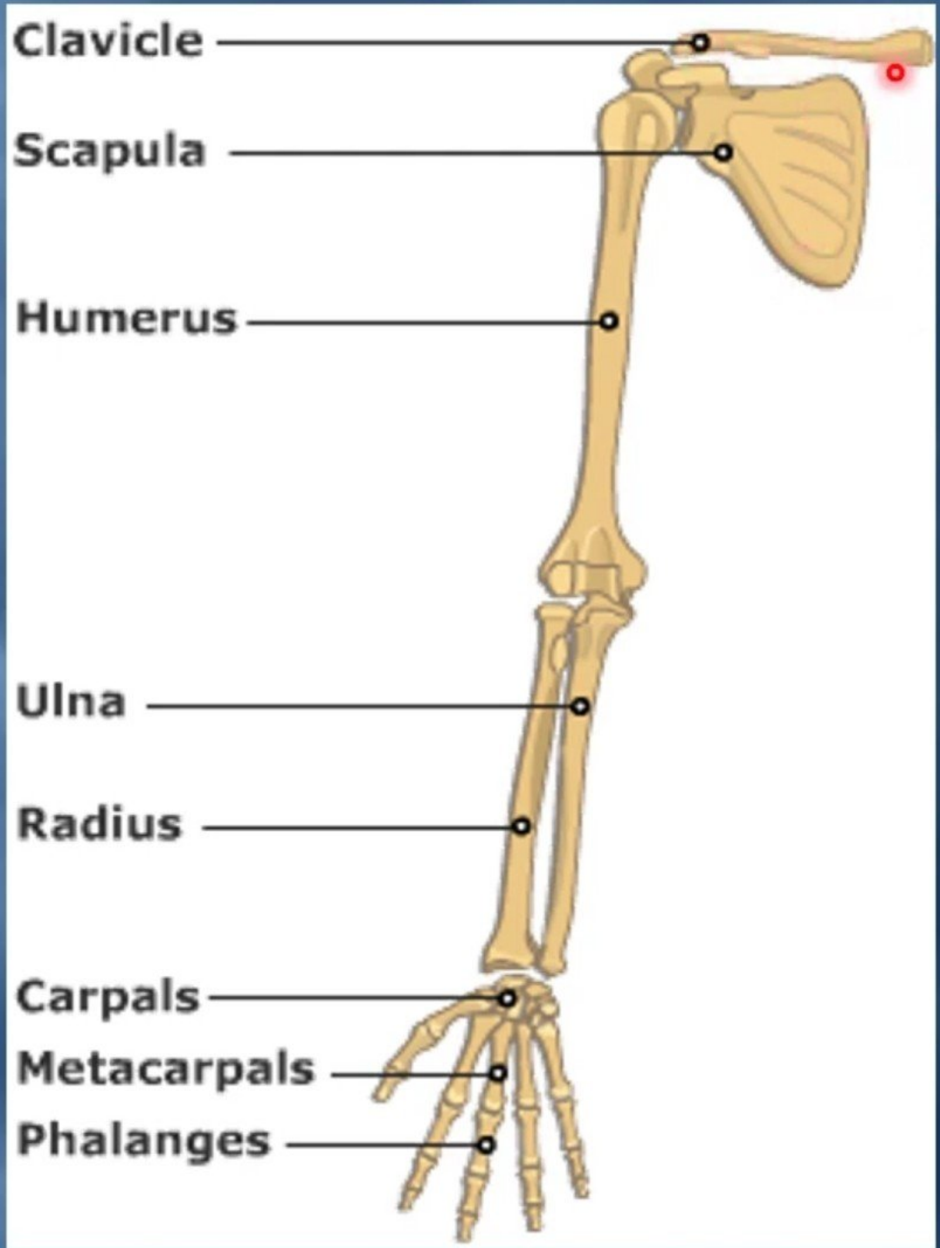




The Upper Limbs

The Bones of Upper Limb are:

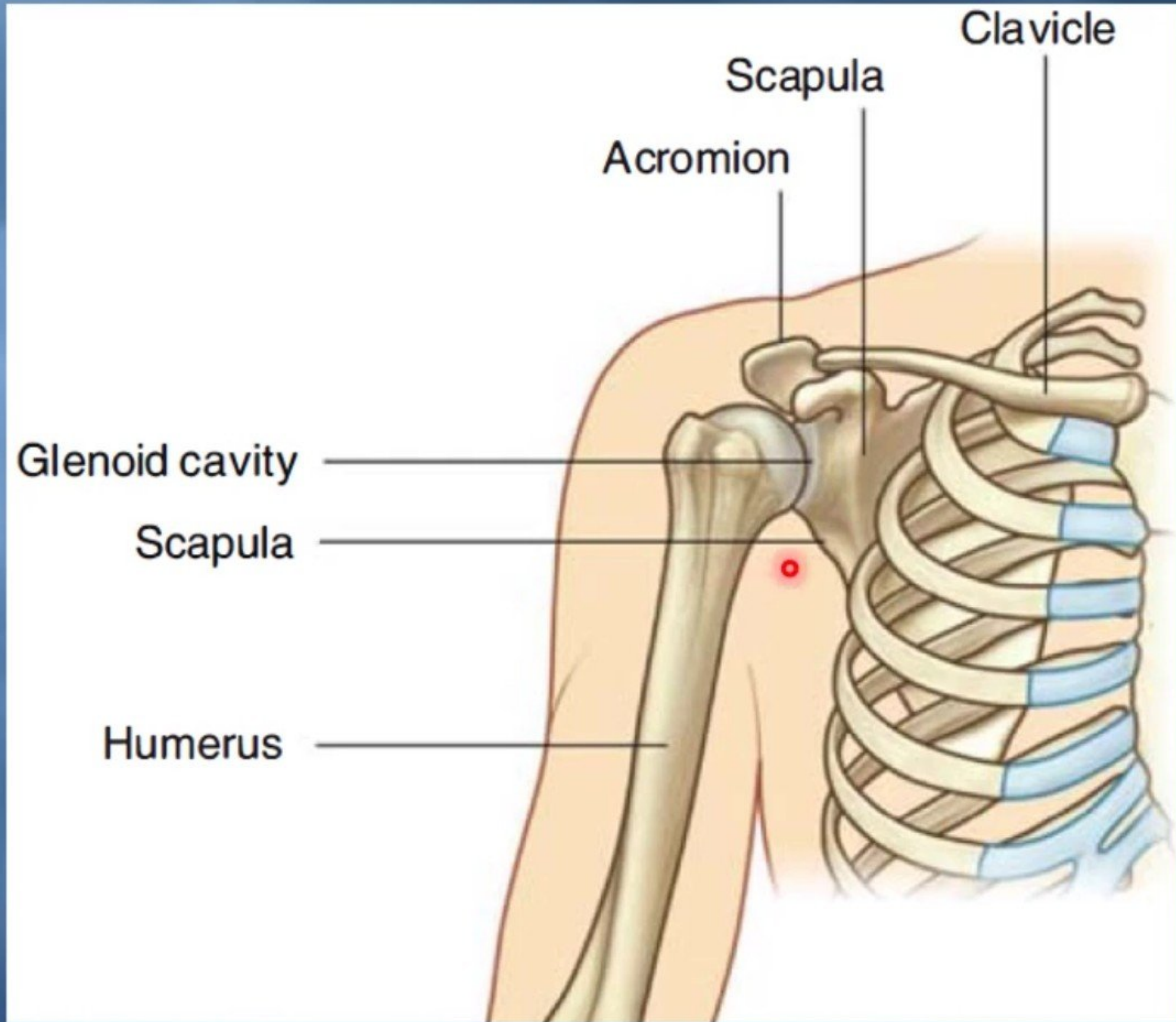
- ✓ **Pectoral girdle or *Shoulder Girdle:*** ***clavicle &scapula***
- ✓ ***Arm :* *Humerus.***
- ✓ ***Forearm :* *Radius & Ulna.***
- ✓ ***Hand:* *Carpal bones, Metacarpal & Phalanges***



Pectoral or Shoulder Girdle Girdle

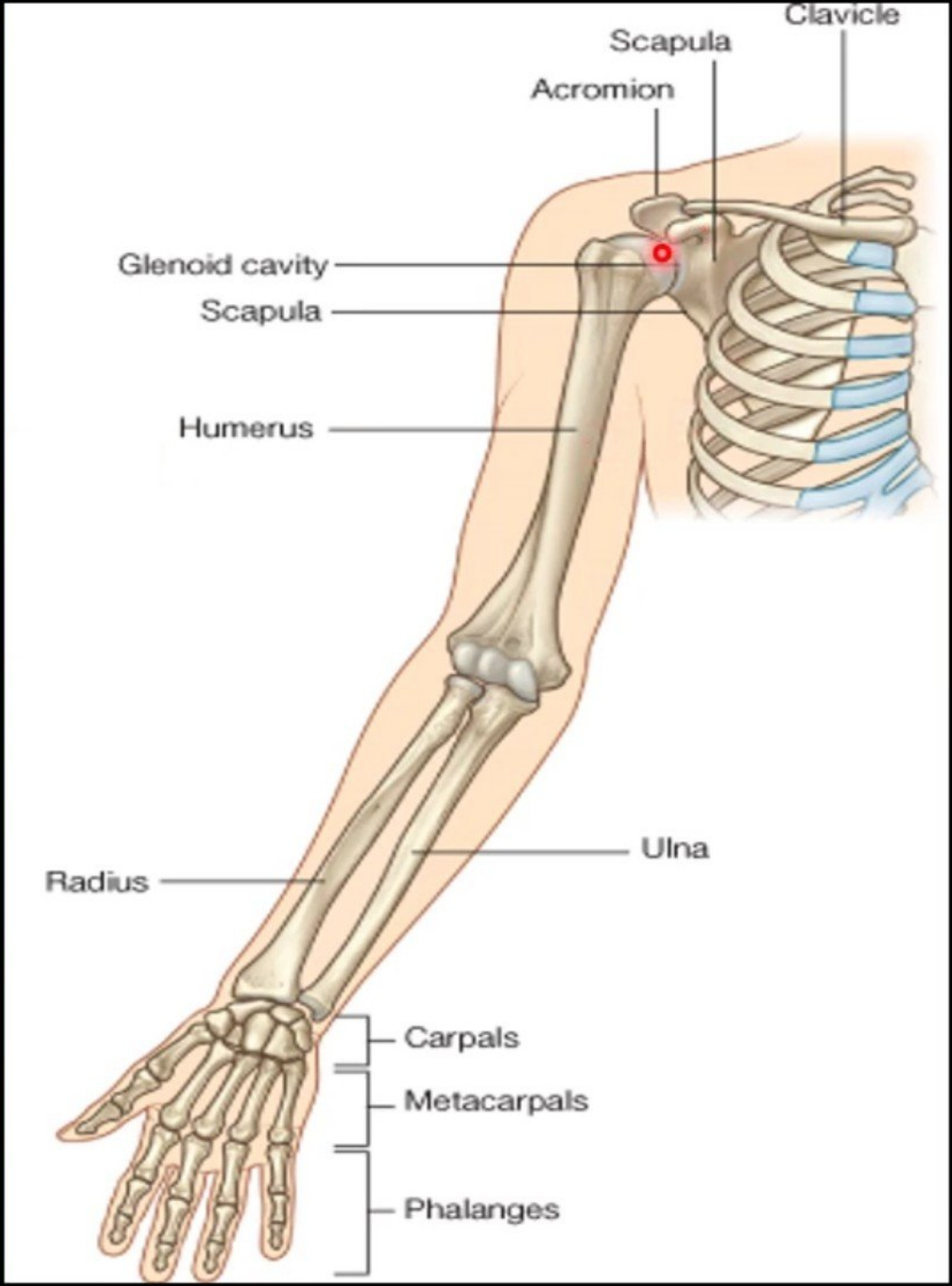
- *Formed of Two Bones:*
- *Clavicle (anteriorly).*
- *Scapula(posteriorly).*

- *It is very light and allows the upper limb to have exceptionally free movement*



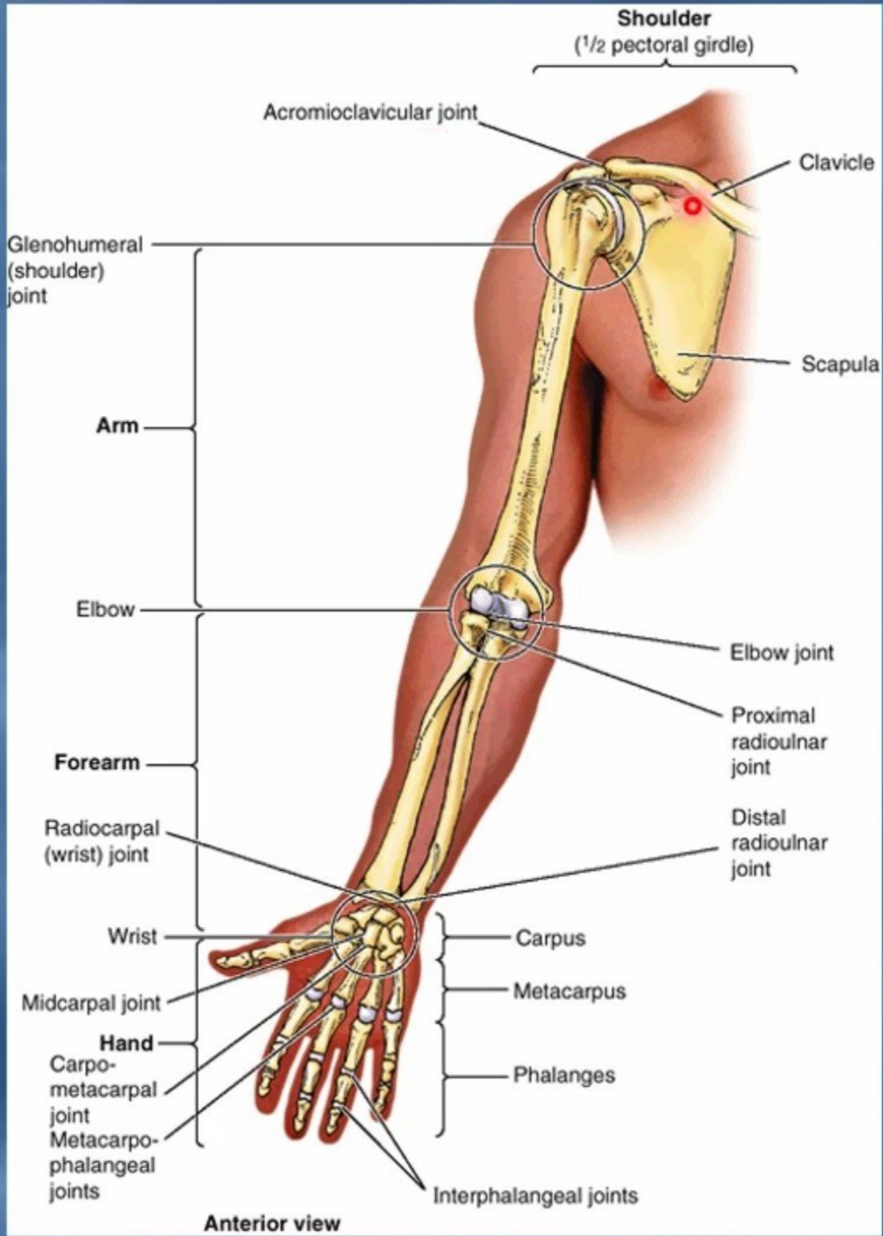
Pectoral or Shoulder Girdle

- The bones of the pectoral or shoulder girdle, the **clavicle** and **scapula**, connect the upper limbs (**humerus**) to the axial skeleton.
- The **sternoclavicular joint**: is only one small joint connects the girdle to the rest of the skeleton.
- The remaining attachment to the axial skeleton is mainly **muscular**.
- clavicle is anchored to **the first costal cartilage** by the **costoclavicular ligament**.
- The **end of the clavicle** normally **transmits much force**.

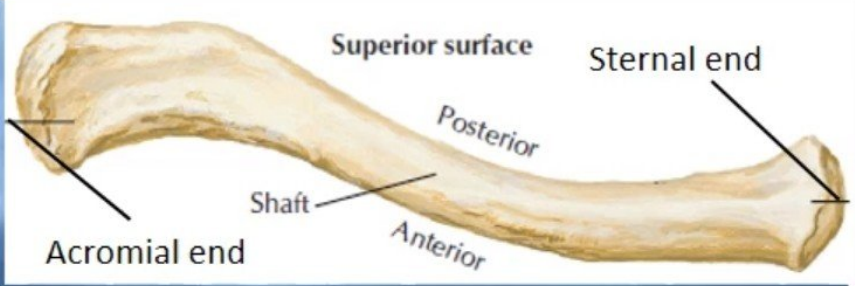


Pectoral or Shoulder Girdle

- Almost all **movement between humerus** and **glenoid cavity** at the **shoulder joint** is accompanied by an appropriate movement of **the scapula** itself.
- The **scapula cannot move without** making its **supporting strut**, **the clavicle**, **move** also.
- Generally speaking the shoulder joint, the **acromioclavicula**r and **sternoclavicular joints** all move together in harmony, providing a kind of '**thoracohumeral articulation**'.
- **Defects in** any part of the '**thoracohumeral articulation**' must **impair the function** of the **whole**.

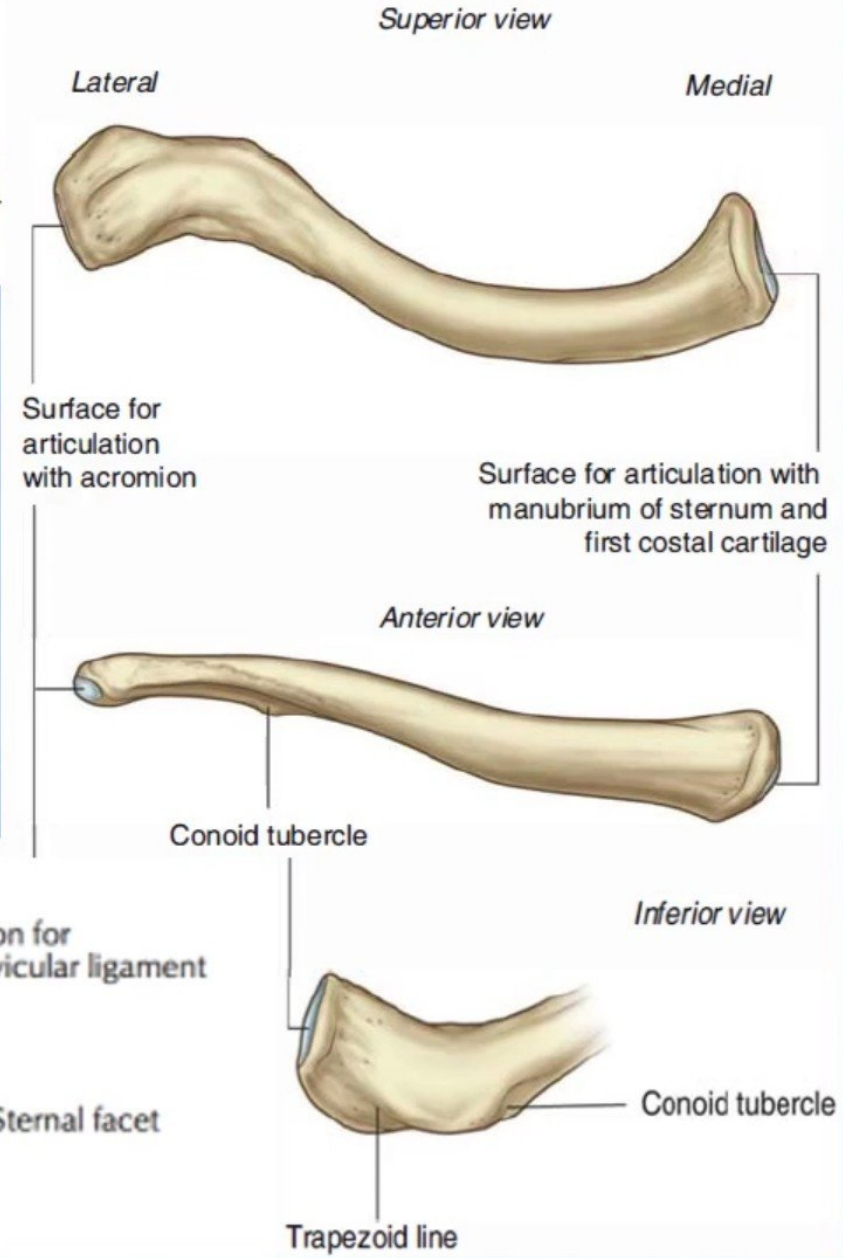


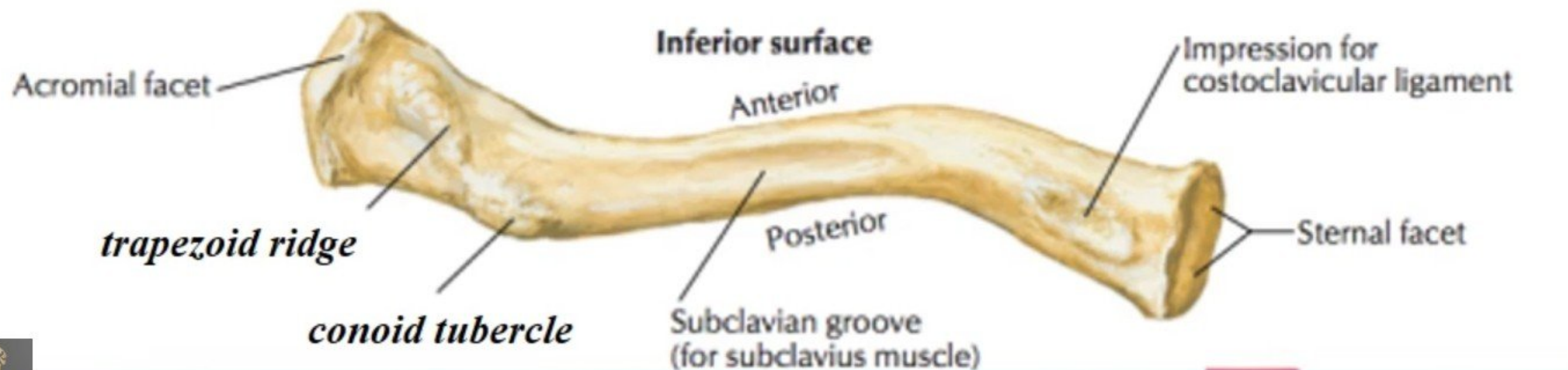
Clavicle



Surfaces:

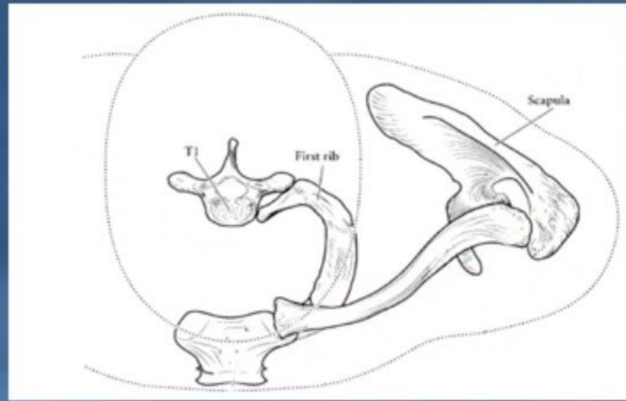
- ***Superior :*** *smooth as it lies just deep to the skin.*
- ***Inferior :*** *rough because strong ligaments bind it to* the 1st rib, ***subclavius groove*** *and* ***conoid tubercle*** *and* **trapezoid ridge.**
- **Is the first bone to begin ossification during fetal development, but it is the last one to complete ossification, at approximately 21 years of age.**

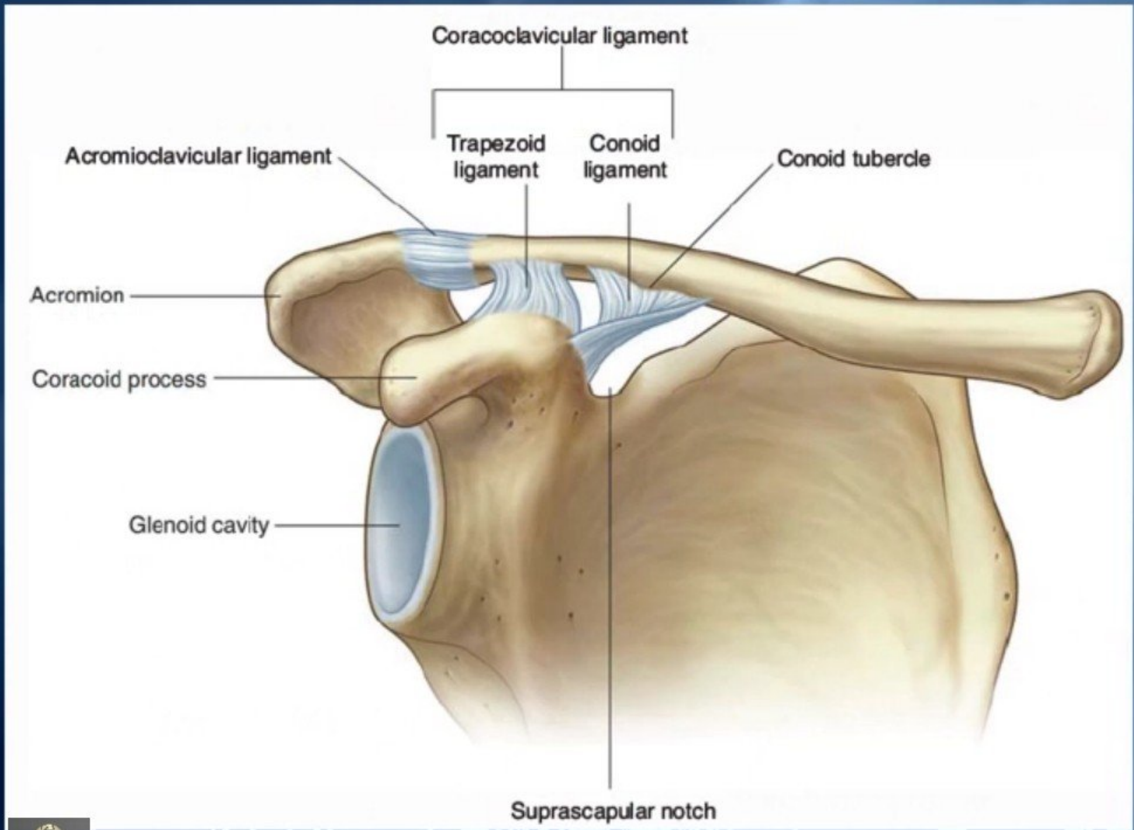


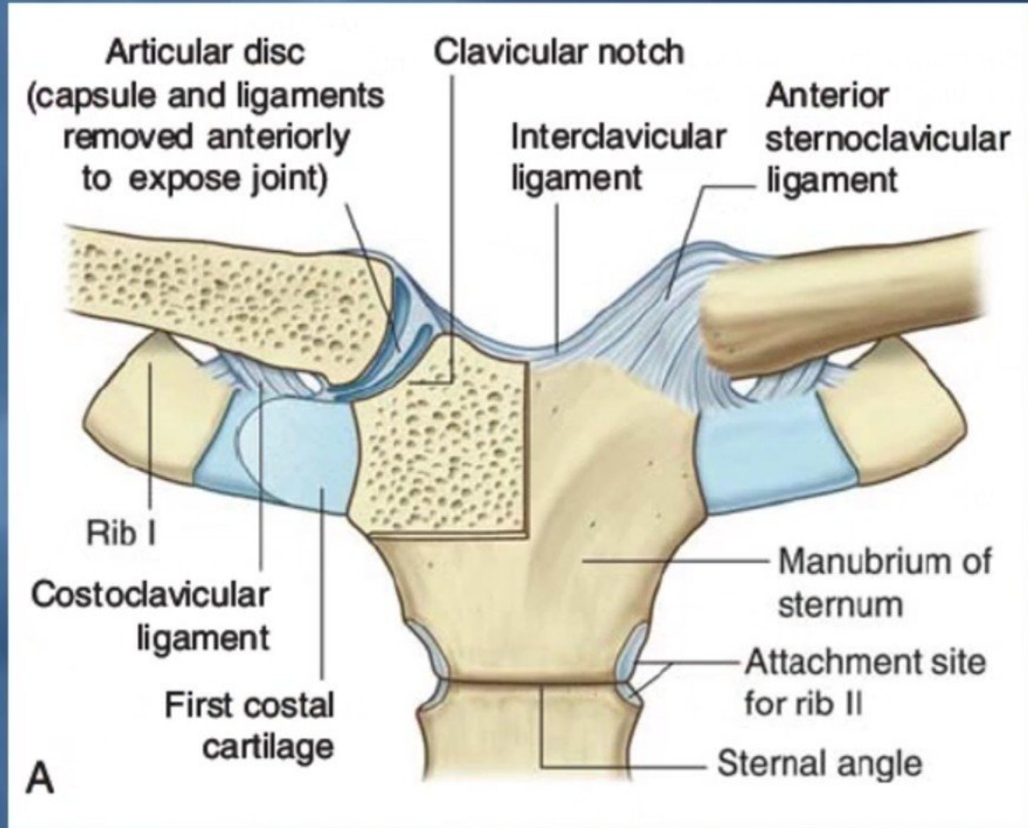


Articulation of the clavice

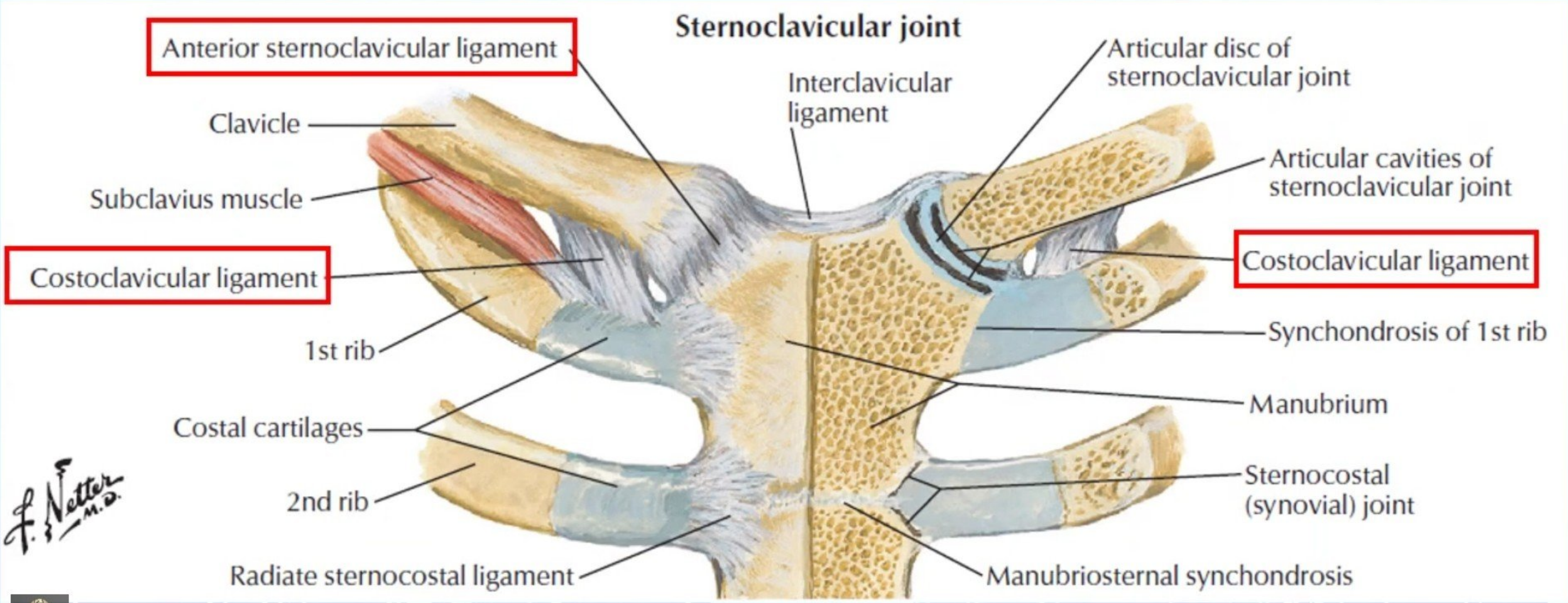
- **<u>Medially</u>** with the manubrium at the ***Sternoclavicular joint*** .
- ***<u>Laterally</u>* with the Scapula at the *Acromioclavicular joint***



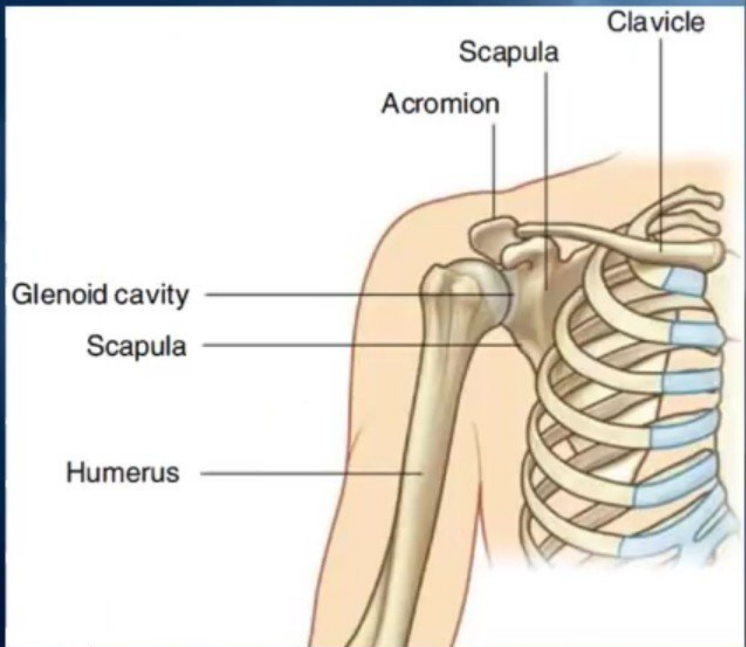


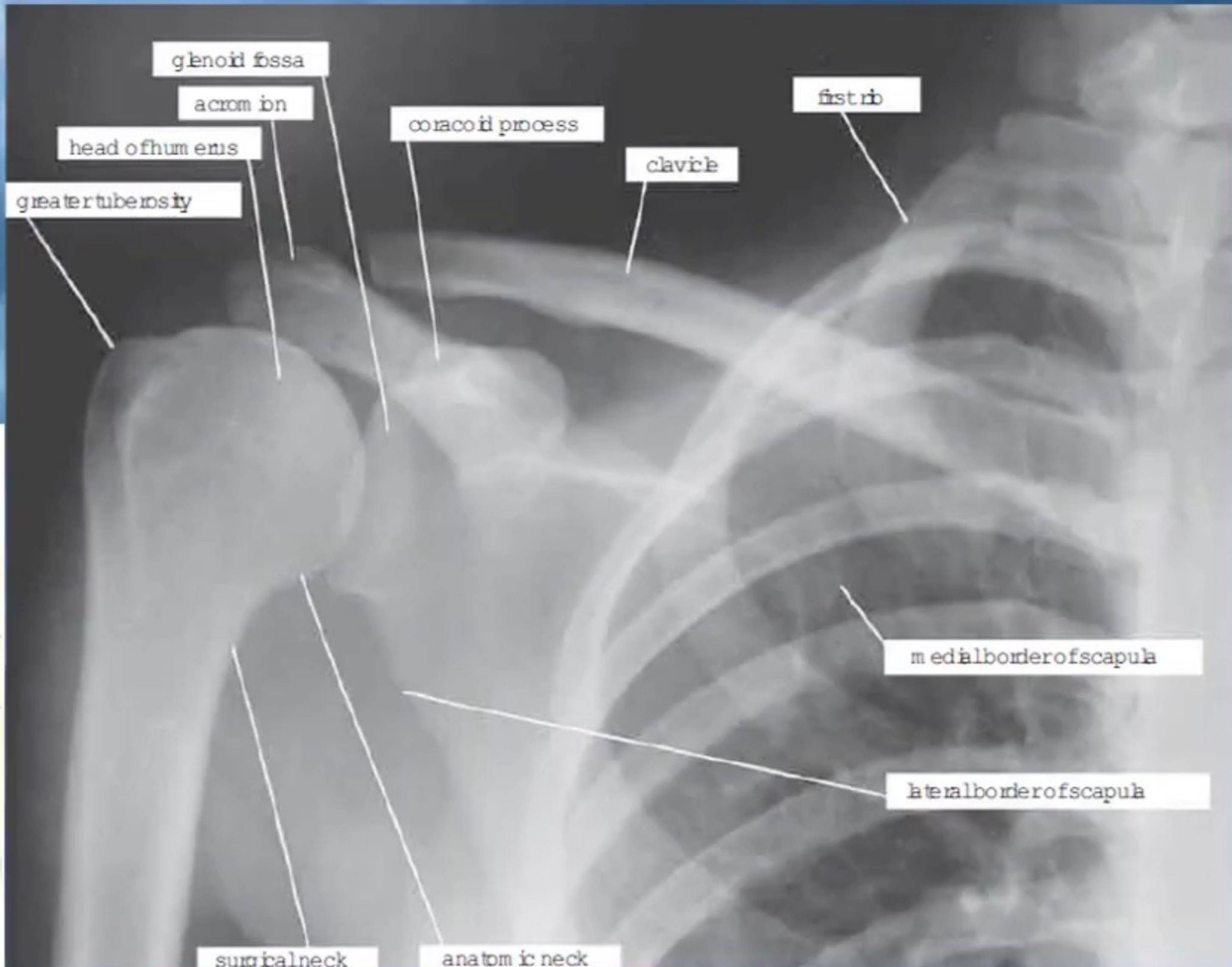


Muscle Origins and Insertions and Ligament attachments



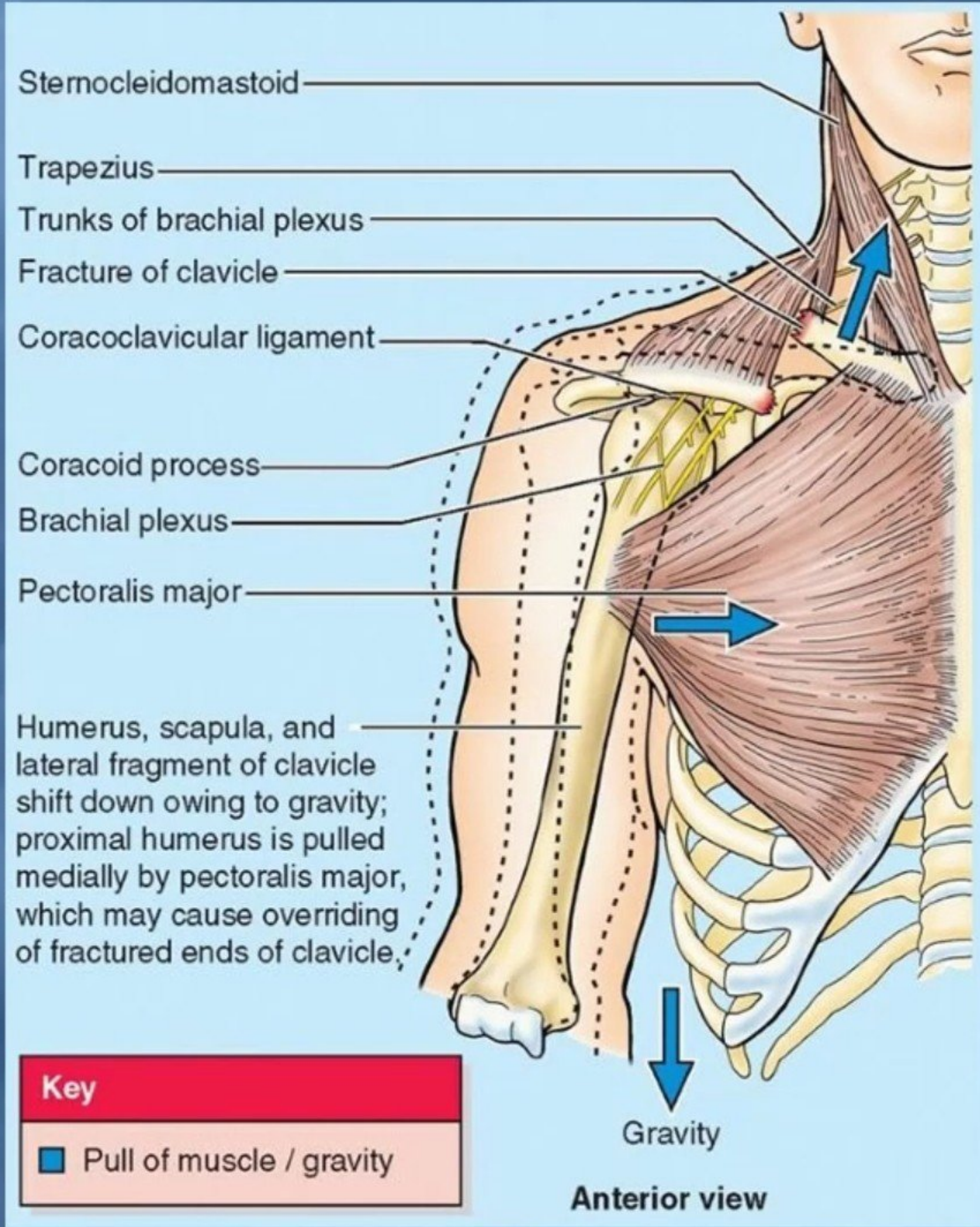
Normal X-ray


Clavicle
Scapula
Acromion
Glenoid cavity
Scapula
Humerus


glenoid fossa
acromion
head of humerus
greater tuberosity
coracoid process
clavicle
first rib
medial border of scapula
lateral border of scapula
surgical neck
anatomic neck

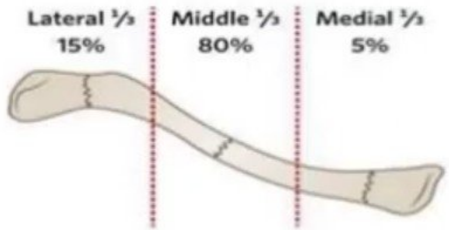
Fractures of the Clavicle

- The clavicle is **commonly fractured** especially in **children** as forces are impacted to the **outstretched hand** during **falling**.
- The **weakest part** of the clavicle is the junction of **the middle and lateral thirds.**
- Fractuer between **the costoclavicular** and **coracoclavicular ligaments** are stronger than the clavicle itself.
- After fracture, **the medial fragment** is **elevated** (by the **sternomastoid muscle**), the **lateral fragment** **drops** because of **the weight of the upper limb.**
- It may be **pulled medially** by the **adductors** of the arm.
- The sagging limb is supported by the other.



Fractures of the Clavicle

- **Classification:**
  - 80% occur in the middle 1/3 (Class A).
  - 15% occur in the lateral or distal 1/3 (Class B).
  - 5% occur in the medial or proximal 1/3 (Class C).




Thanks